

Measurement of π^0 and η mesons in oxygen-oxygen collisions at
 $\sqrt{s_{\text{NN}}} = 5.36$ TeV using Machine Learning-based photon
identification in ALICE Run 3

Supplemental figures

Machine-learning training features in oxygen-oxygen collisions at
 $\sqrt{s_{\text{NN}}} = 5.36$ TeV

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1 Feature map

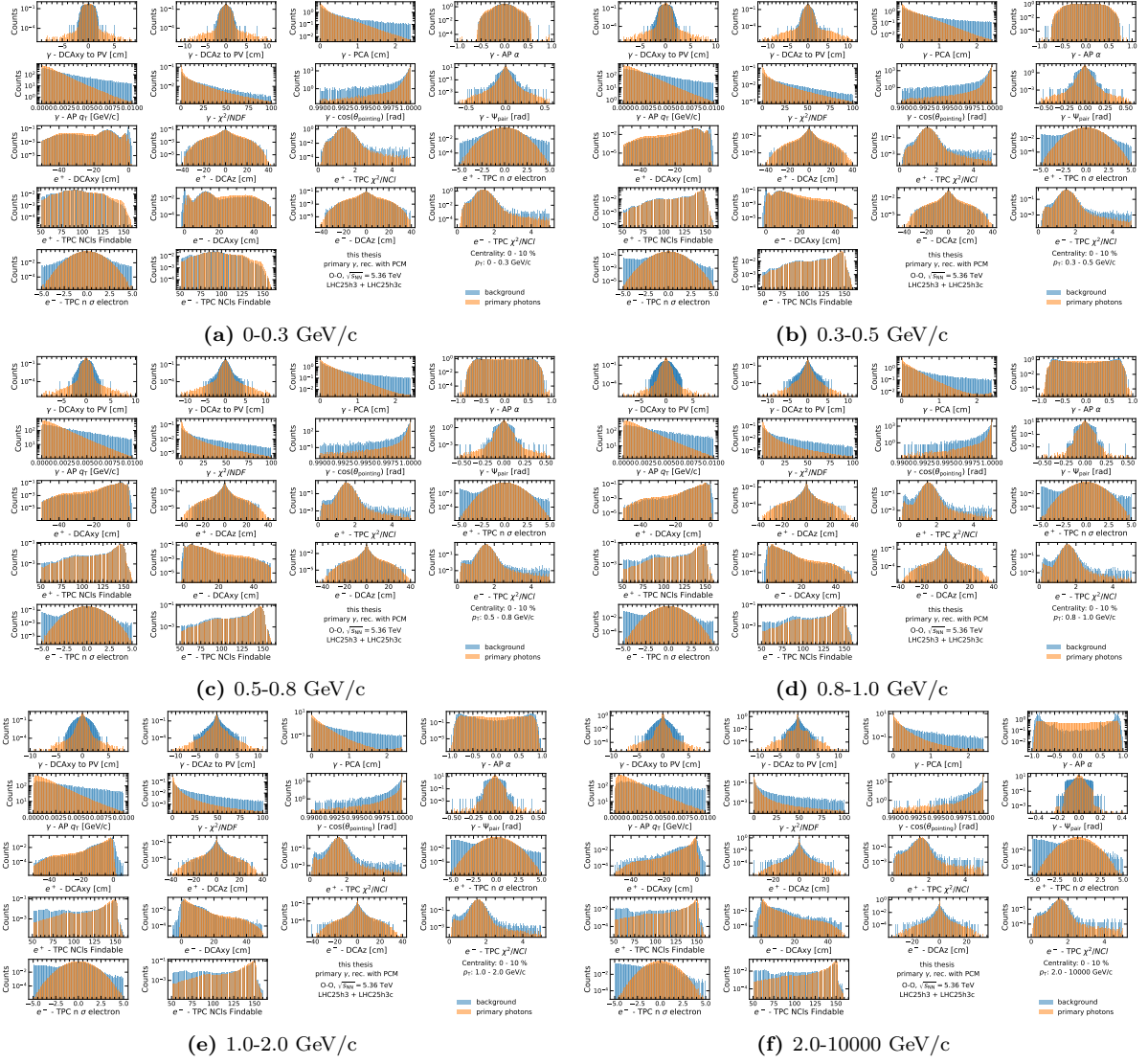


Figure 1: Feature Map for all p_T bins in the centrality range 0-10 %

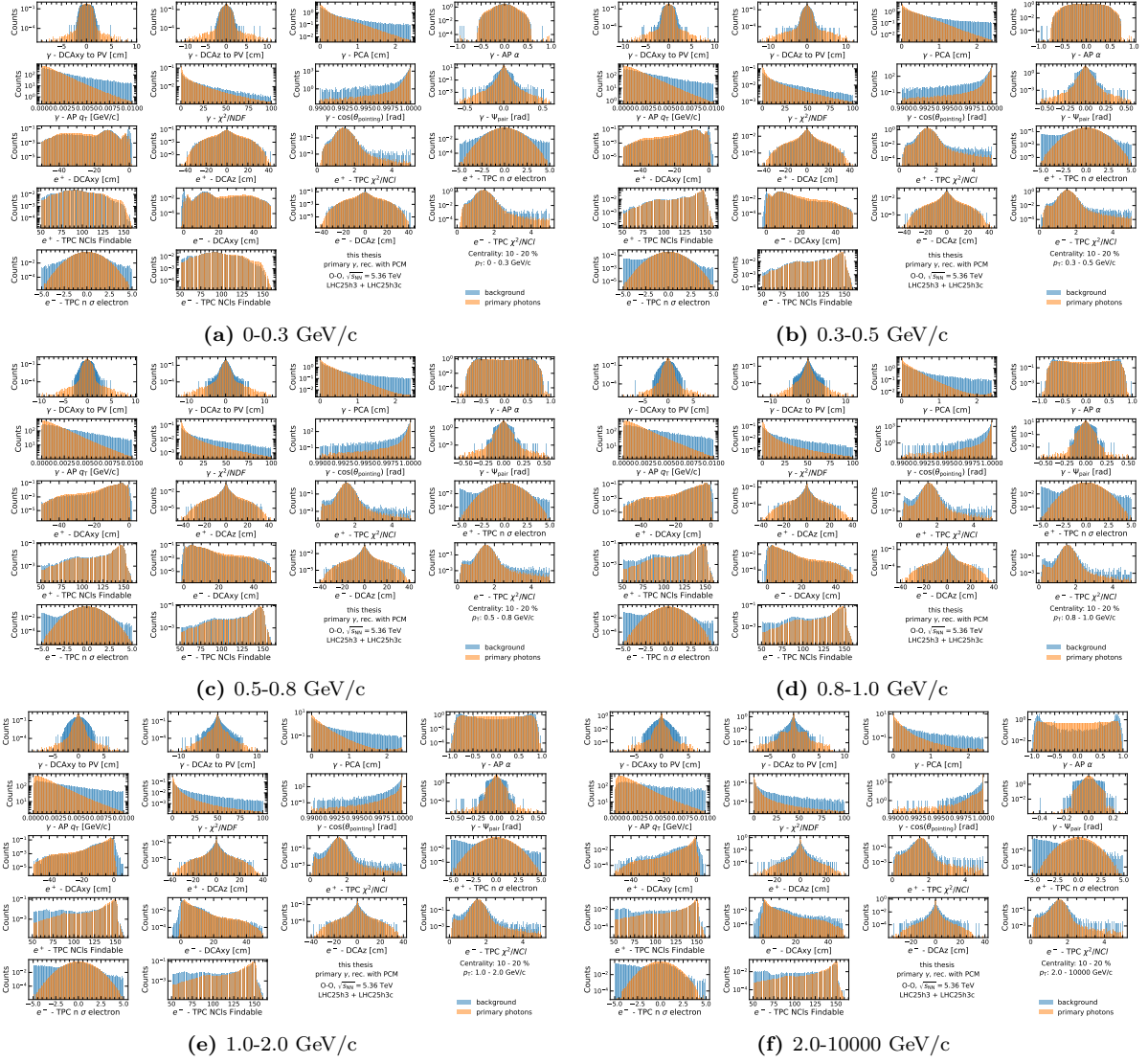


Figure 2: Feature Map for all p_T bins in the centrality range 10-20 %

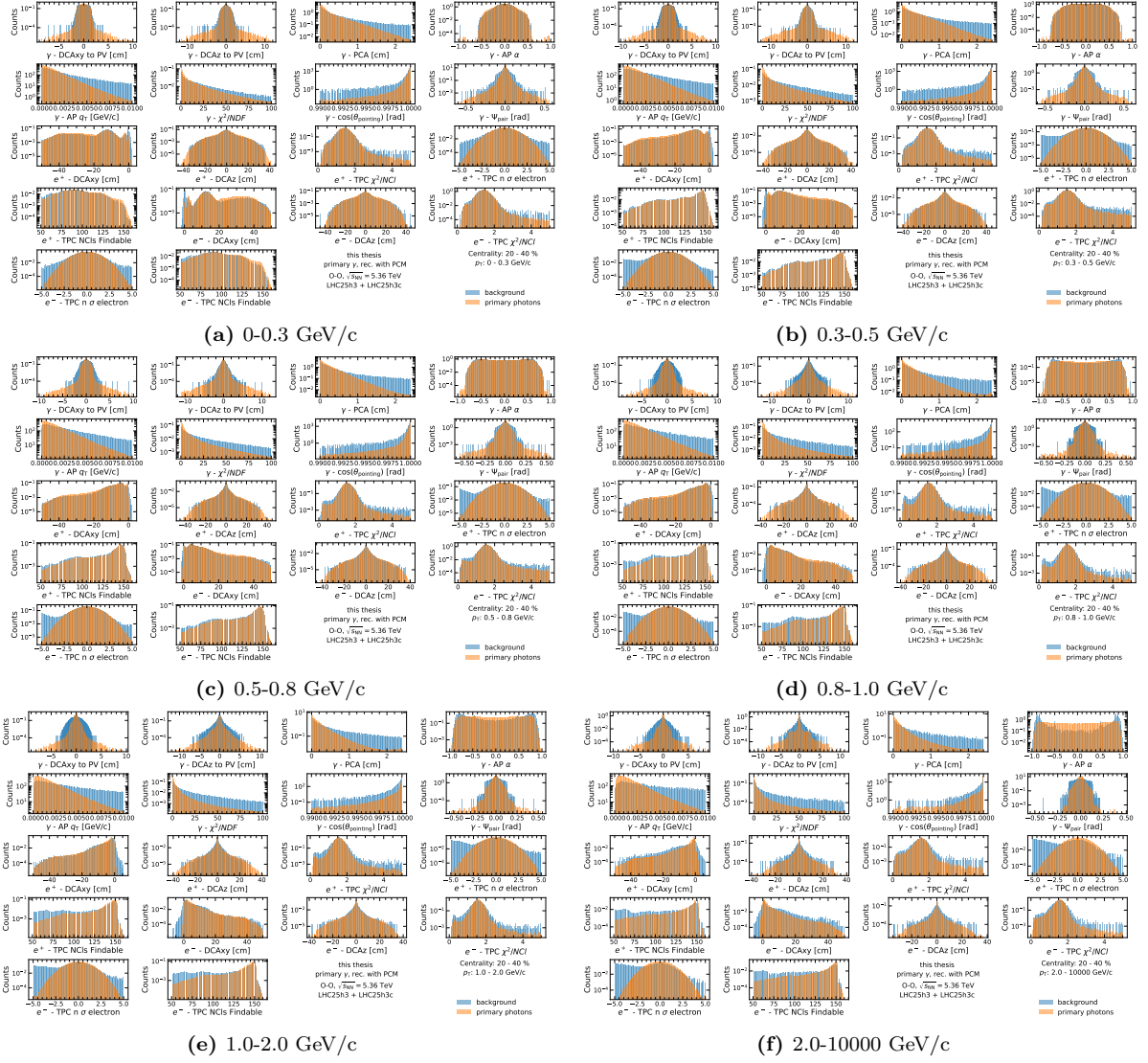


Figure 3: Feature Map for all p_T bins in the centrality range 20-40 %

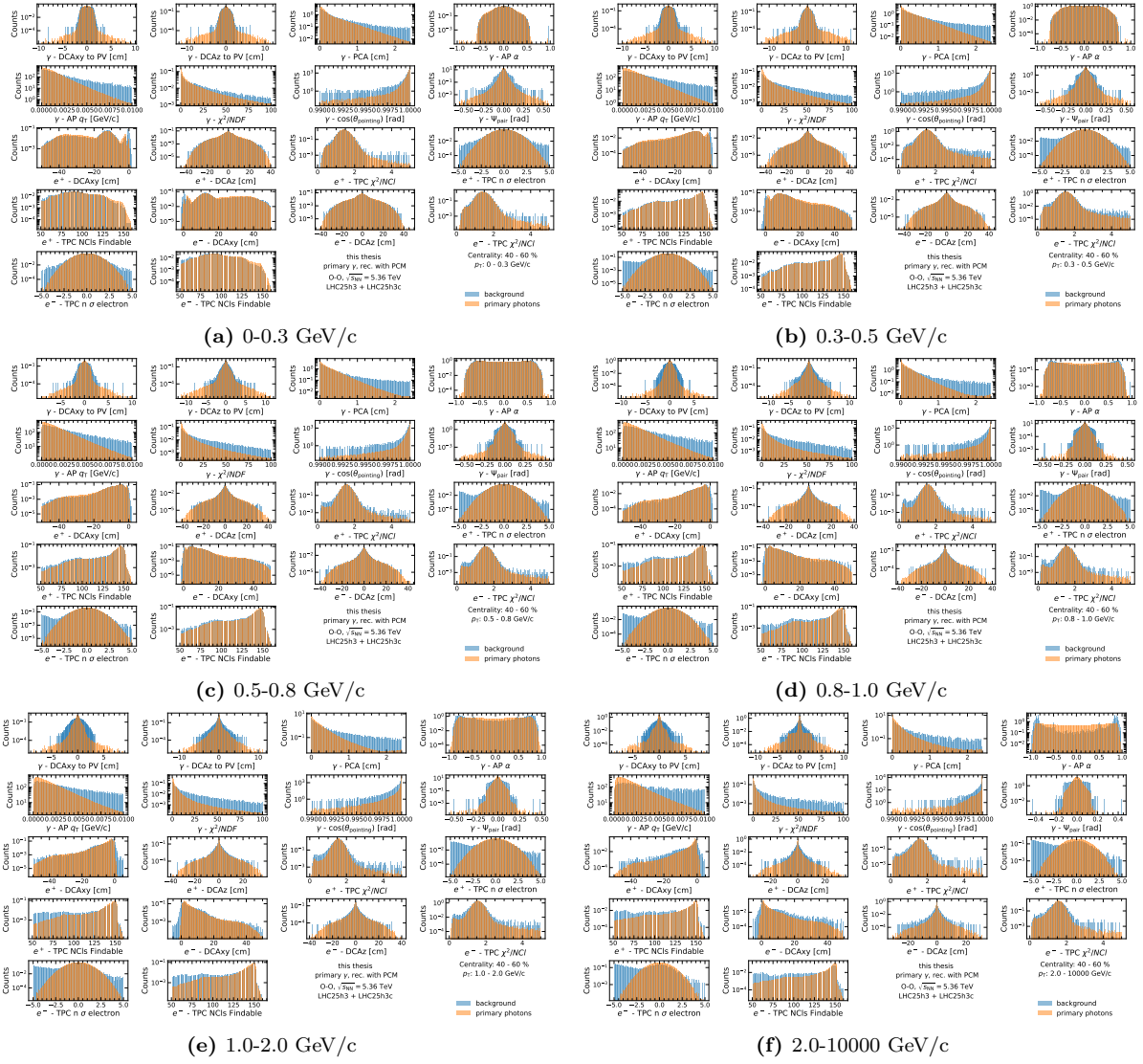


Figure 4: Feature Map for all p_T bins in the centrality range 40-60 %

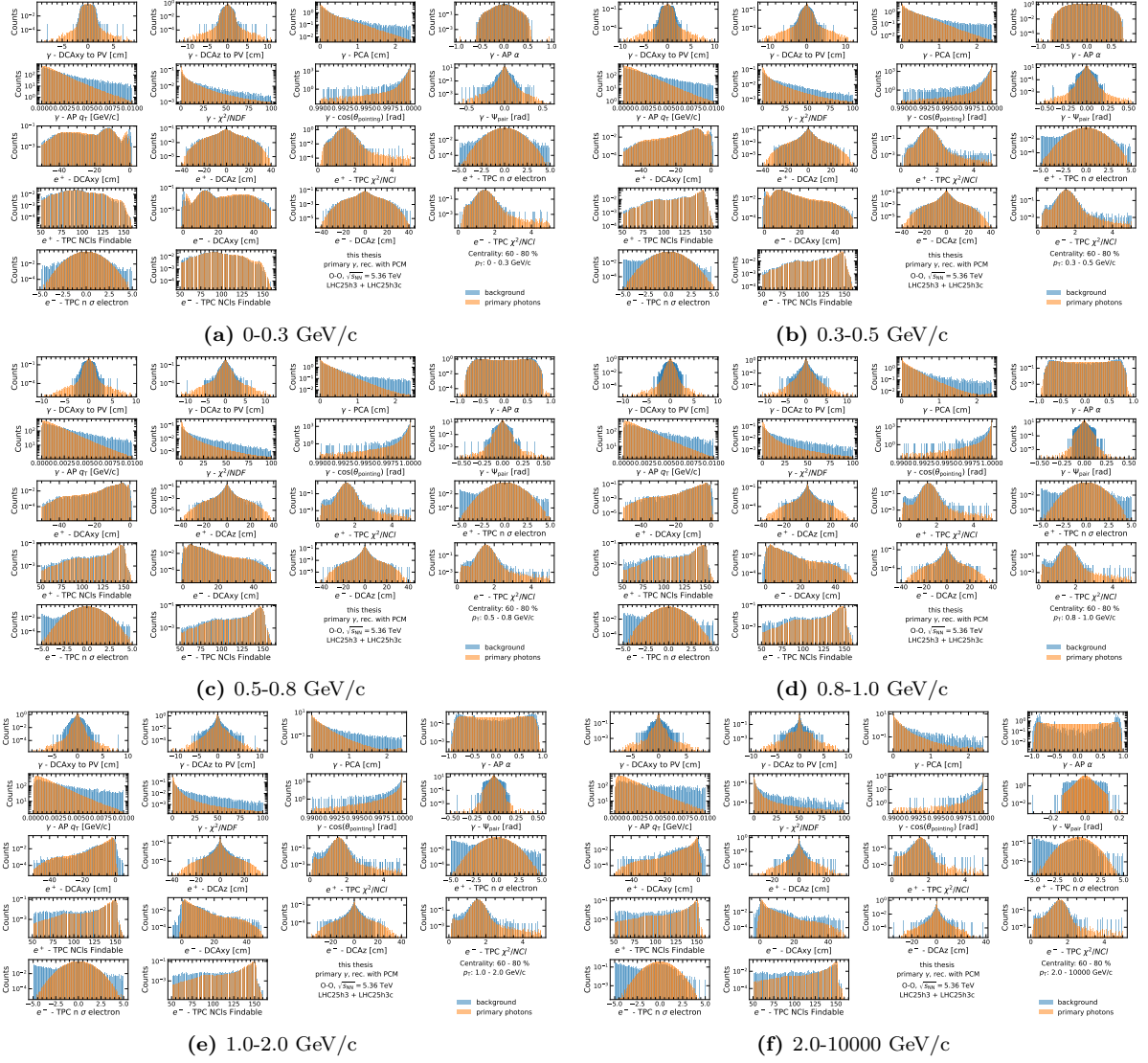


Figure 5: Feature Map for all p_T bins in the centrality range 60-80 %

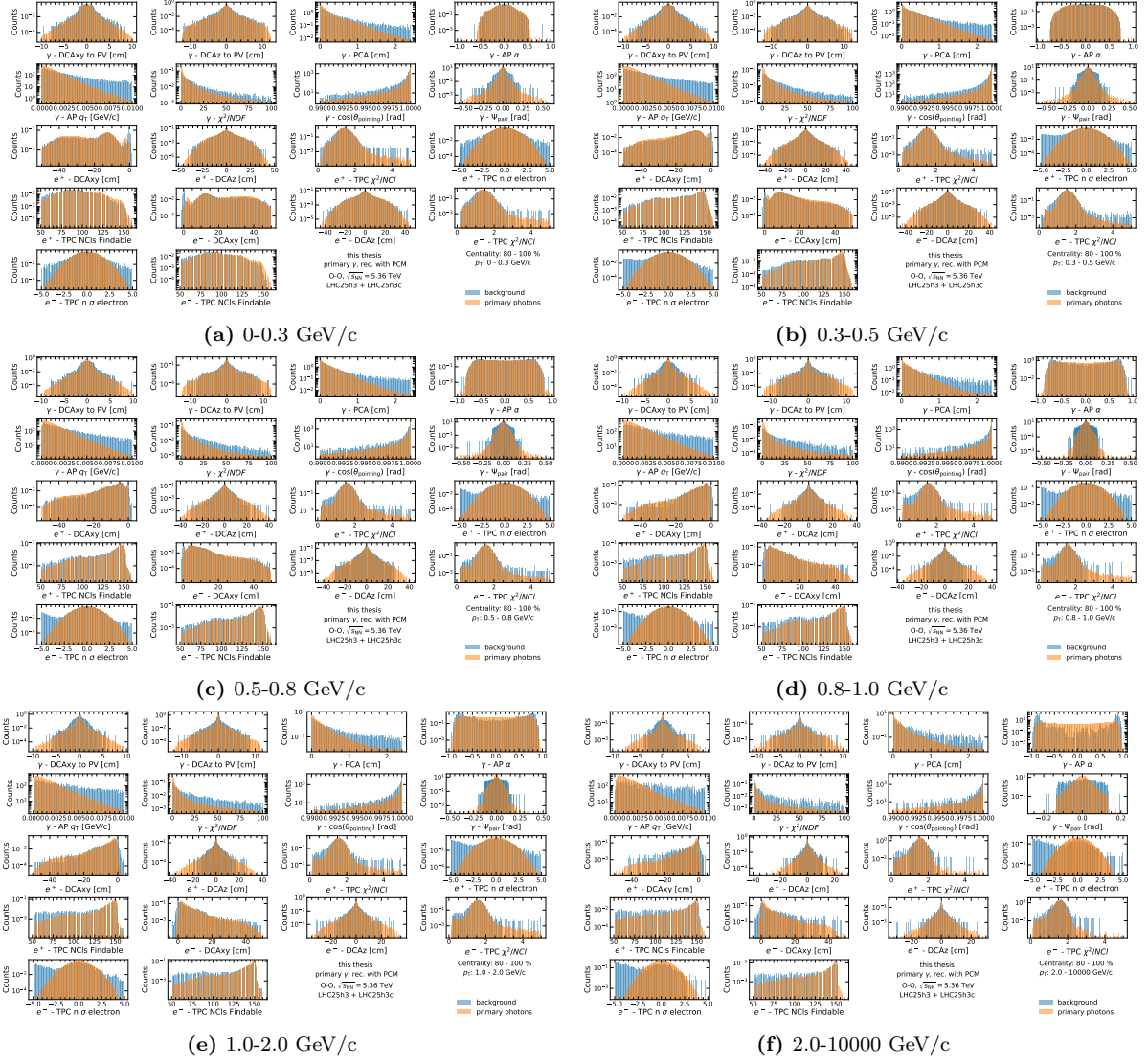


Figure 6: Feature Map for all p_T bins in the centrality range 80-100 %

2 Importance of the input features

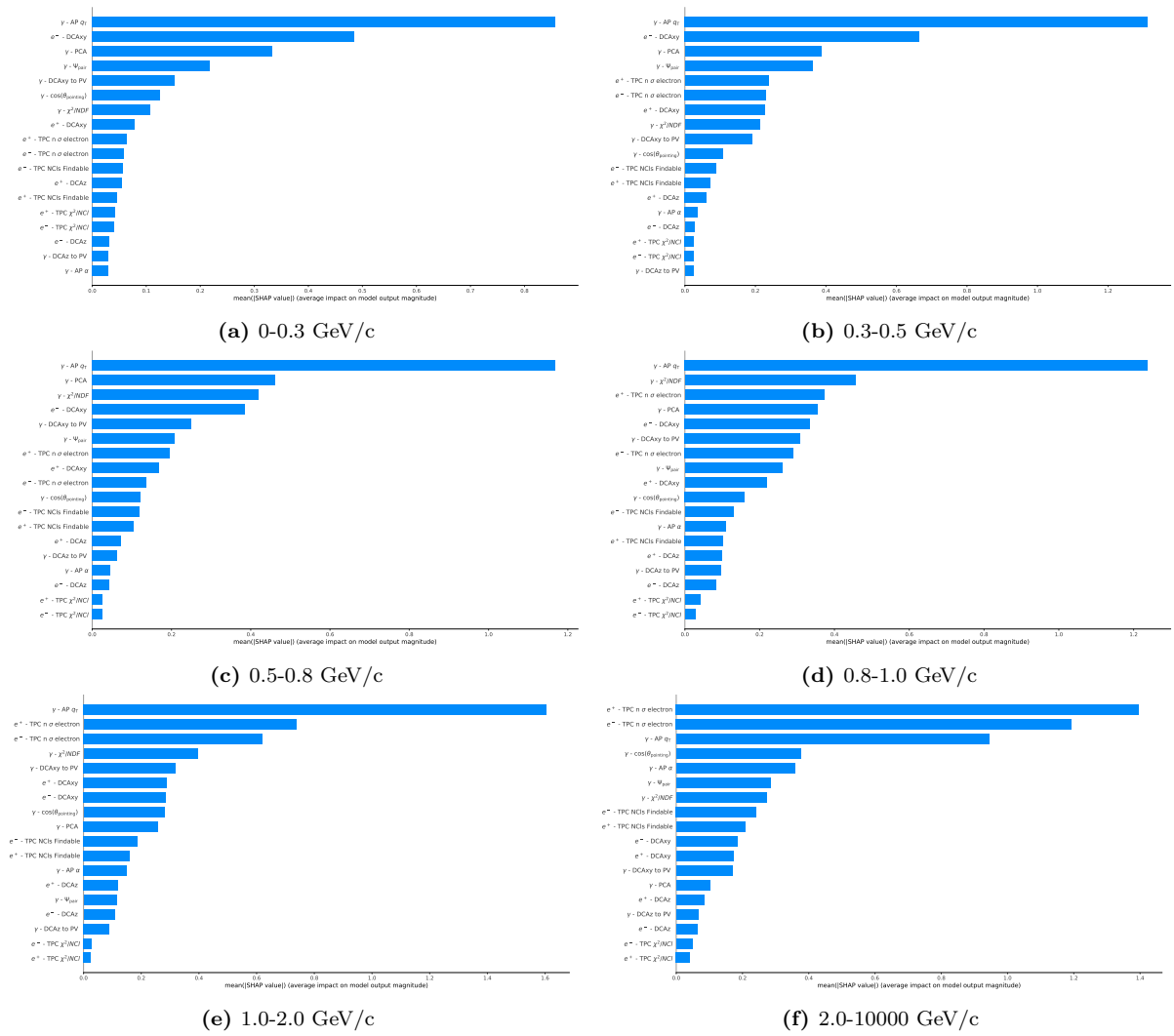


Figure 7: Importance of the input features for all p_T bins in the centrality range 0-10 %

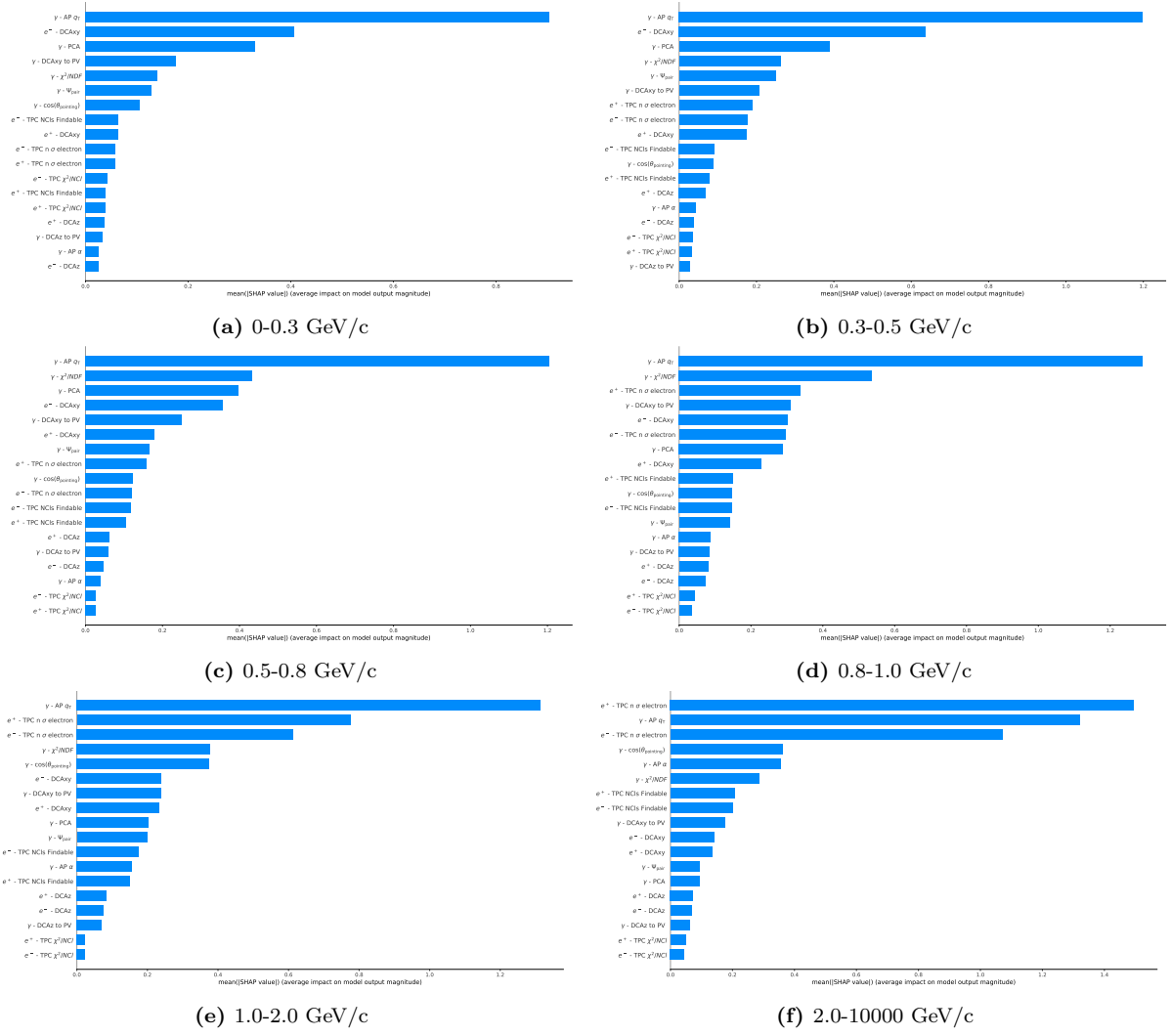


Figure 8: Importance of the input features for all p_T bins in the centrality range 10-20 %

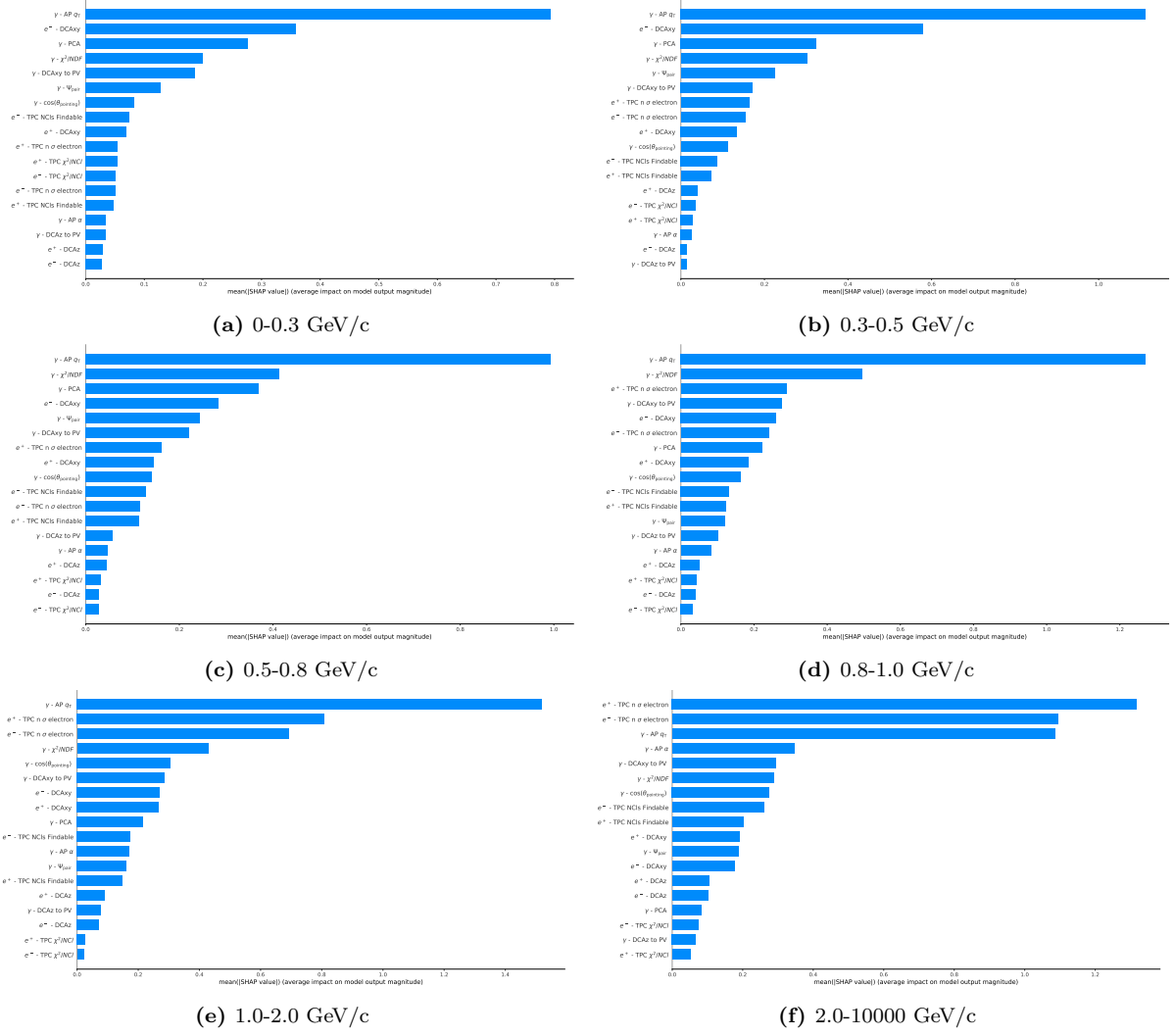


Figure 9: Importance of the input features for all p_T bins in the centrality range 20-40 %

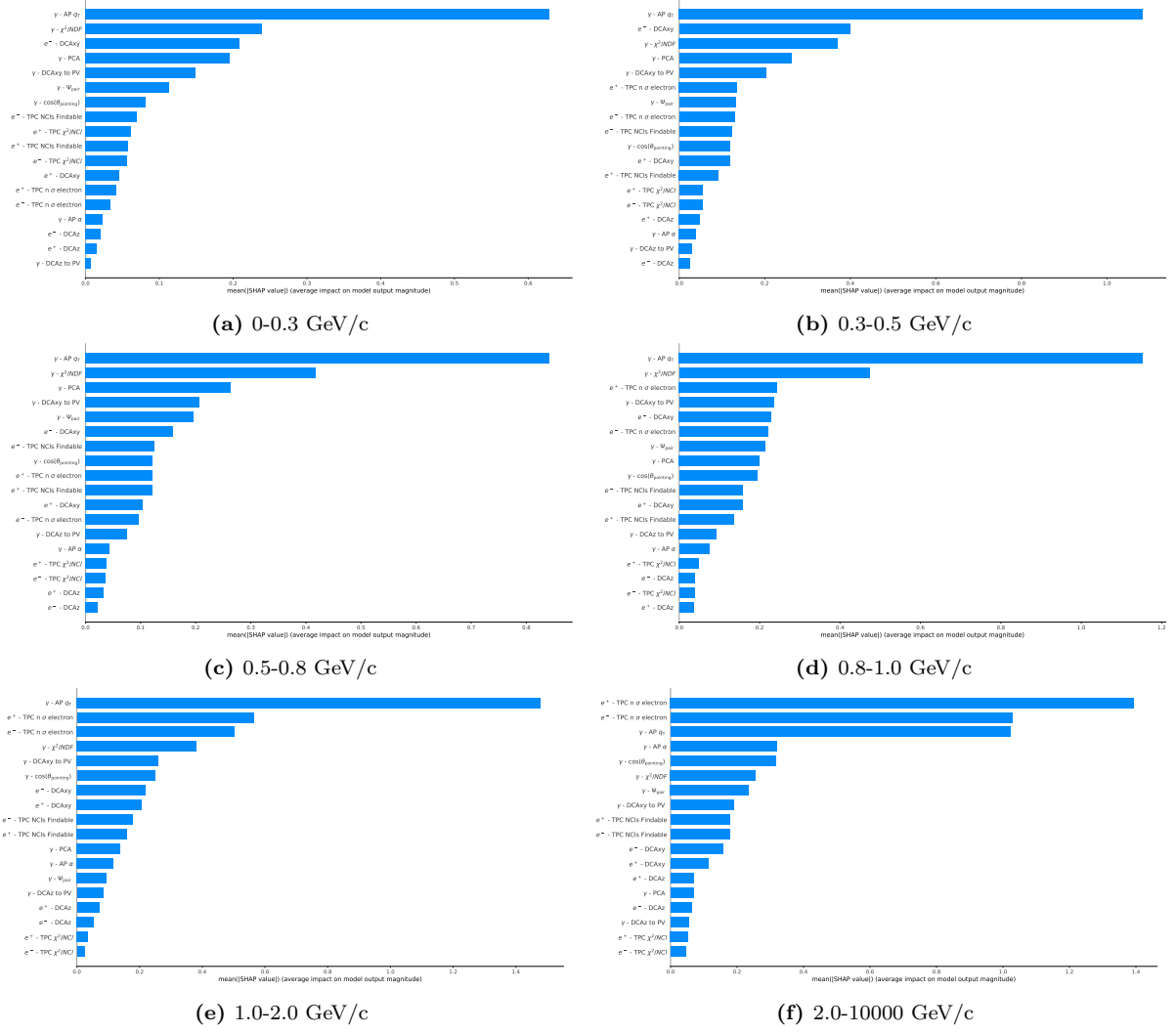


Figure 10: Importance of the input features for all p_T bins in the centrality range 40-60 %

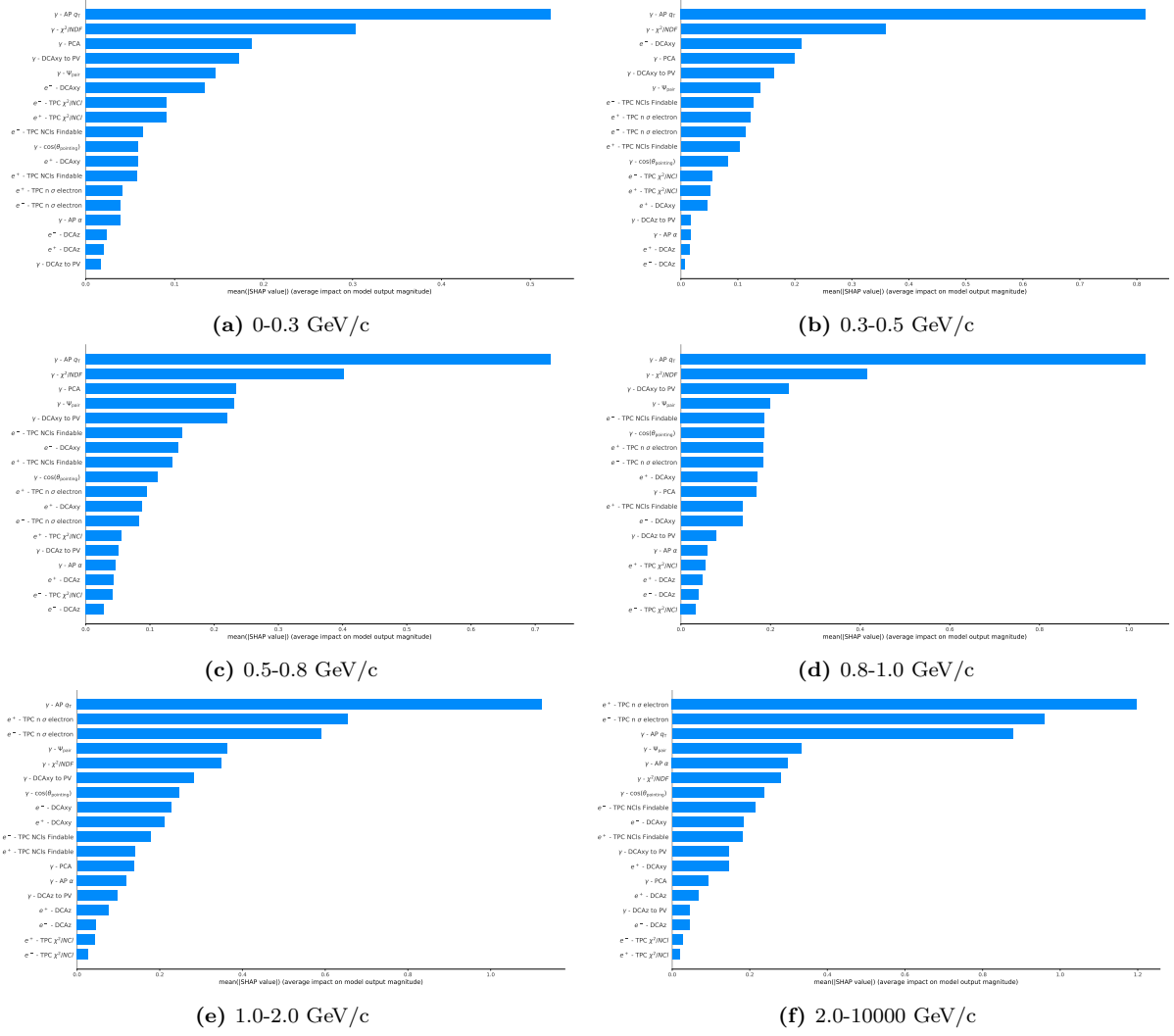


Figure 11: Importance of the input features for all p_T bins in the centrality range 60-80 %

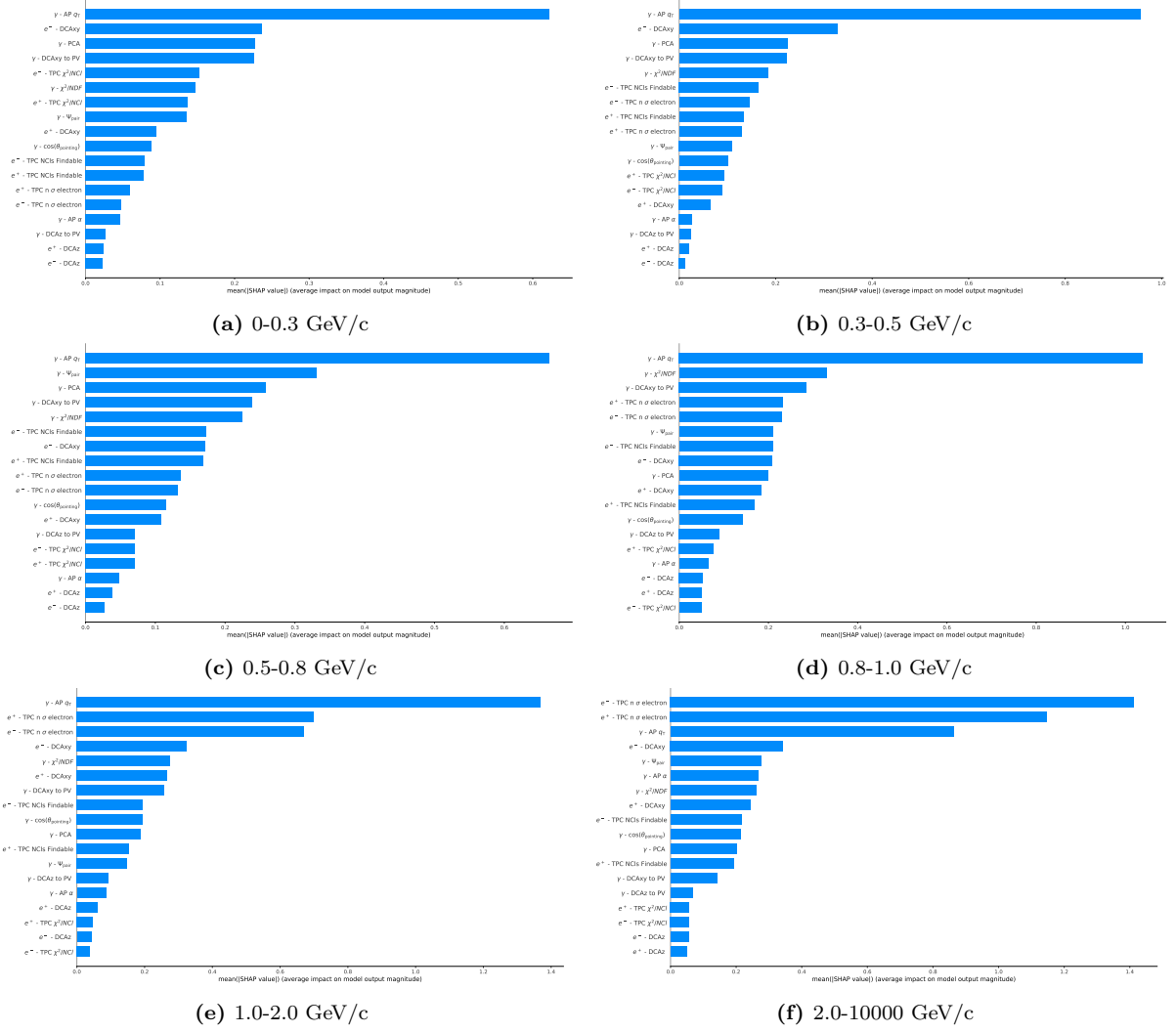


Figure 12: Importance of the input features for all p_T bins in the centrality range 80-100 %

3 Correlation map of the input features

3.1 Correlations of primary photons

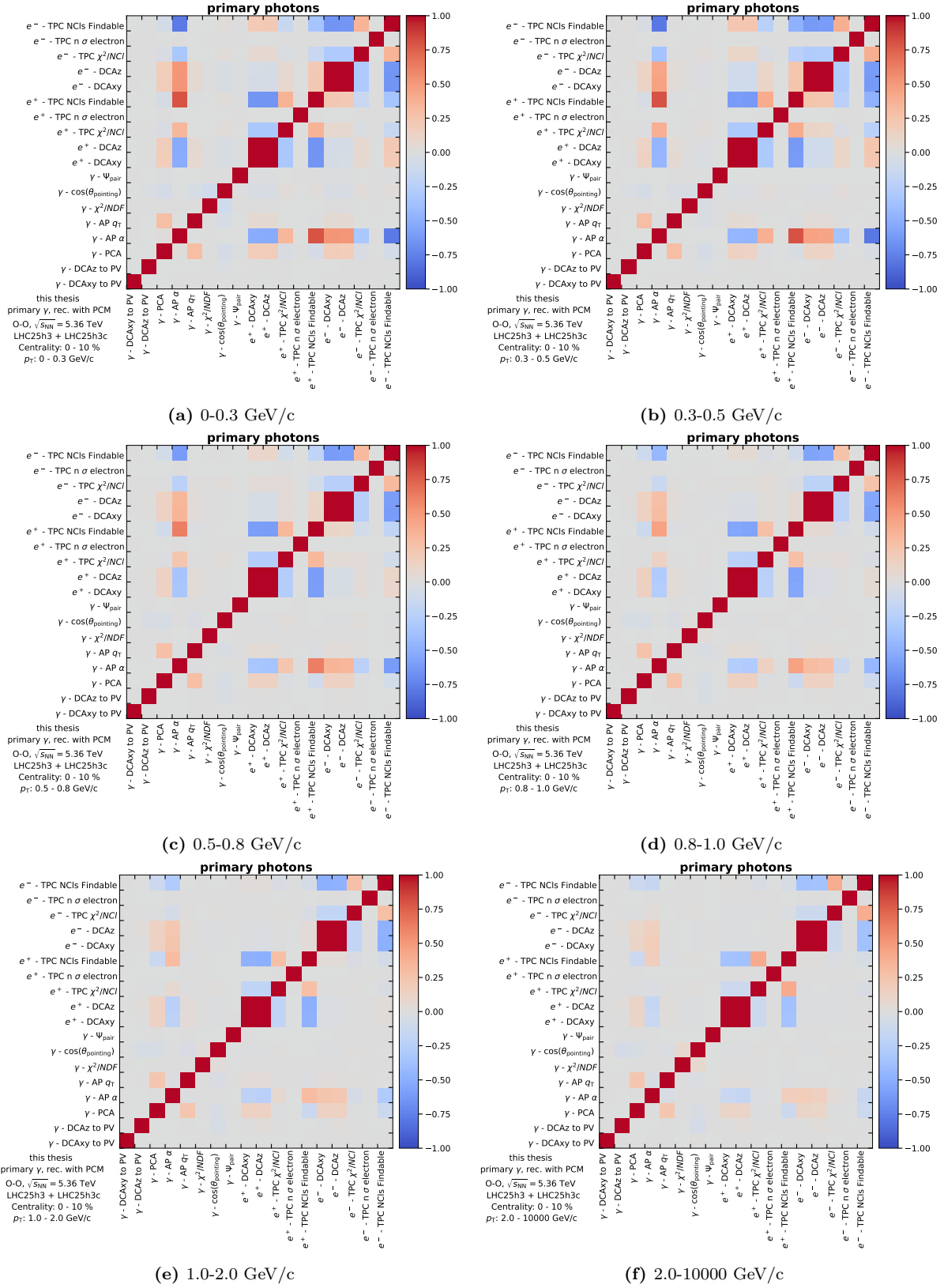
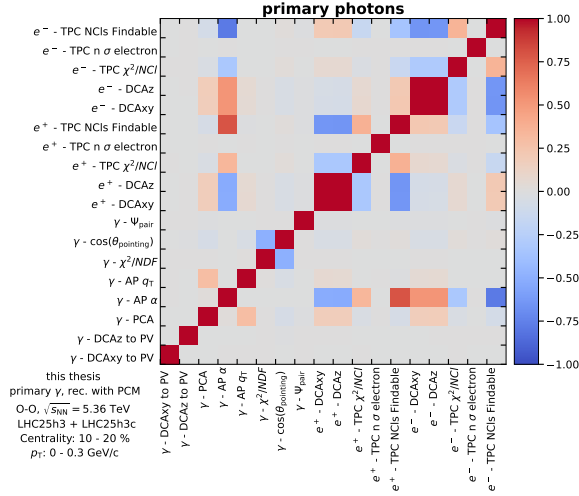
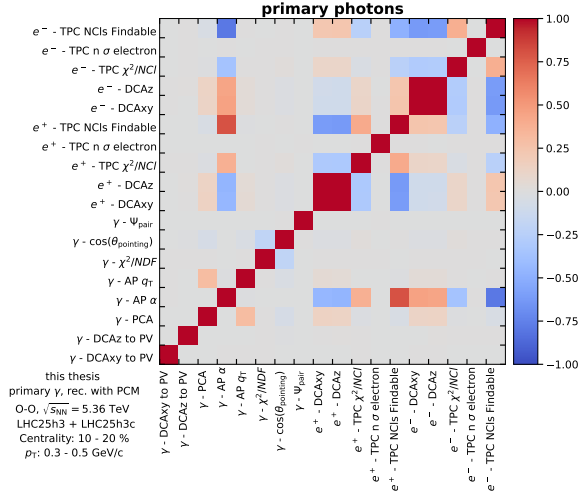


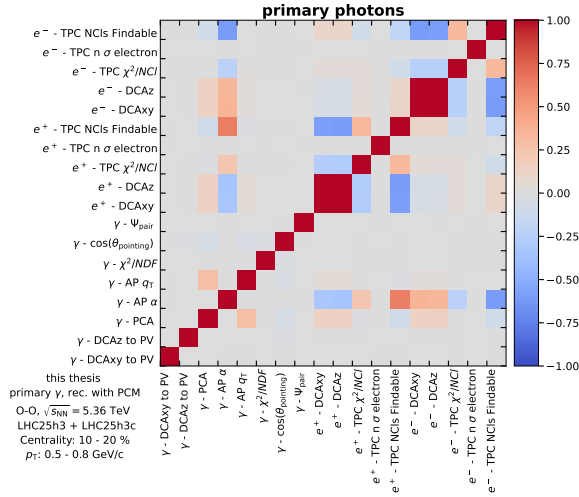
Figure 13: Correlation map of primary photons for all p_T bins in the centrality range 0-10 %



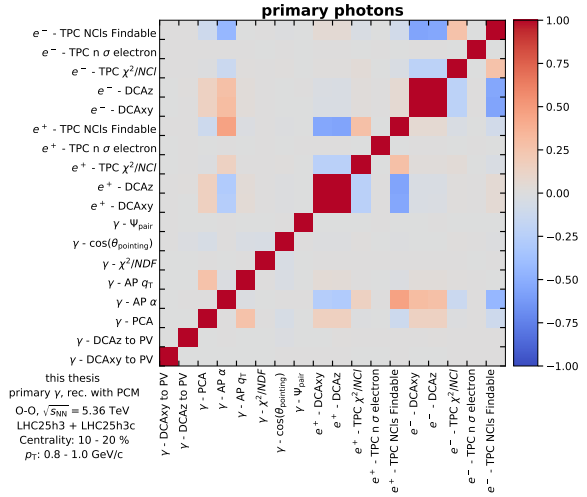
(a) 0-0.3 GeV/c



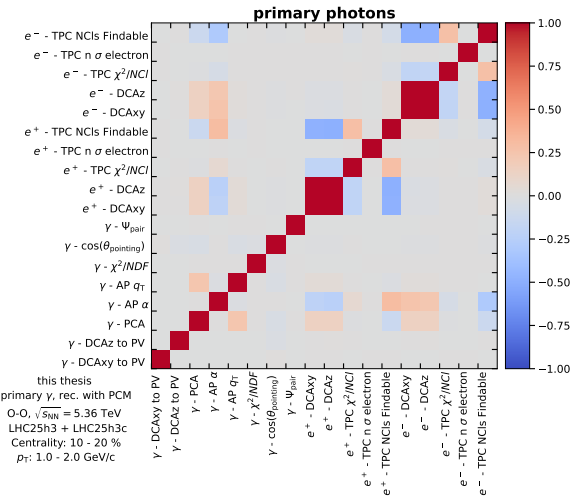
(b) 0.3-0.5 GeV/c



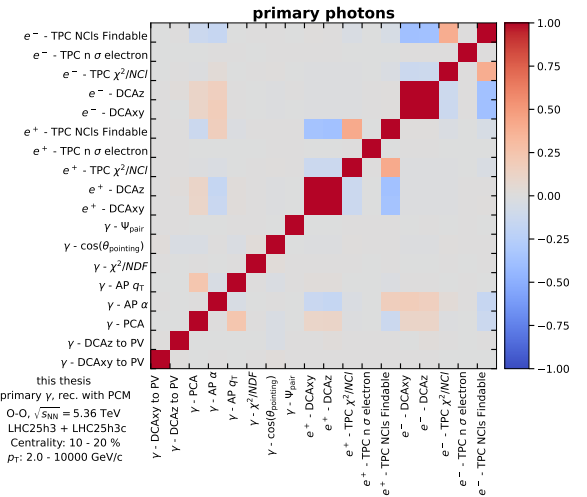
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

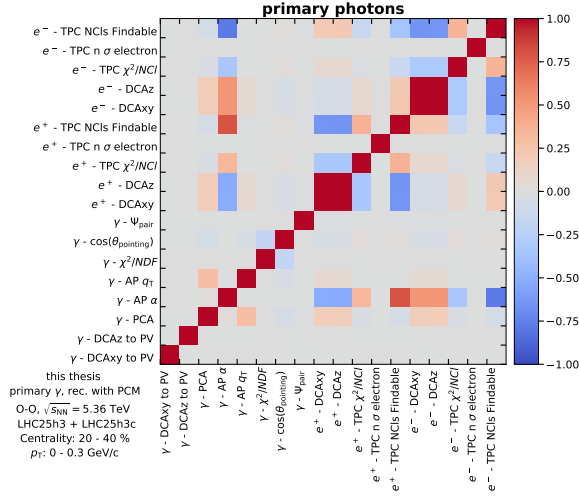


(e) 1.0-2.0 GeV/c

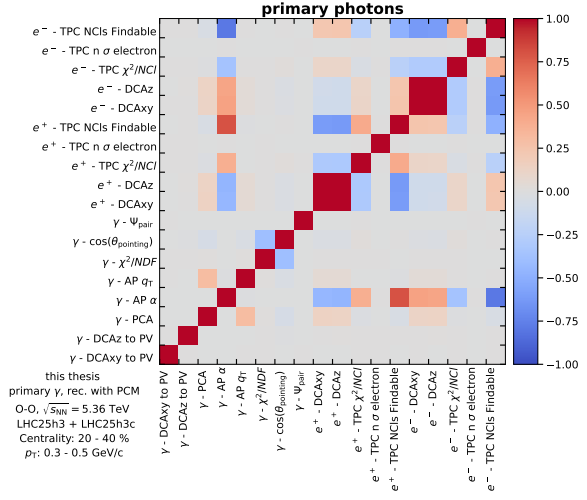


(f) 2.0-10000 GeV/c

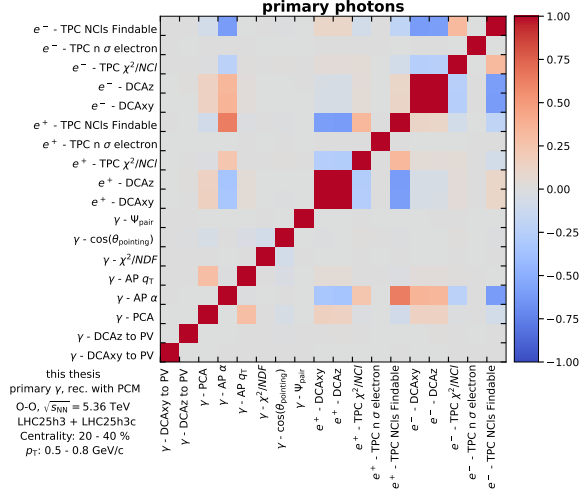
Figure 14: Correlation map of primary photons for all p_T bins in the centrality range 10-20 %



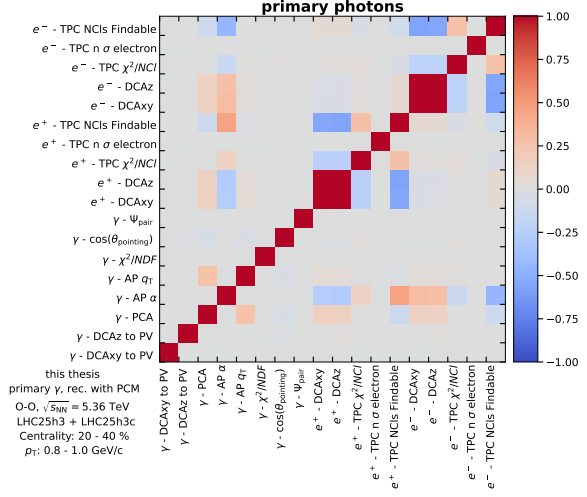
(a) 0-0.3 GeV/c



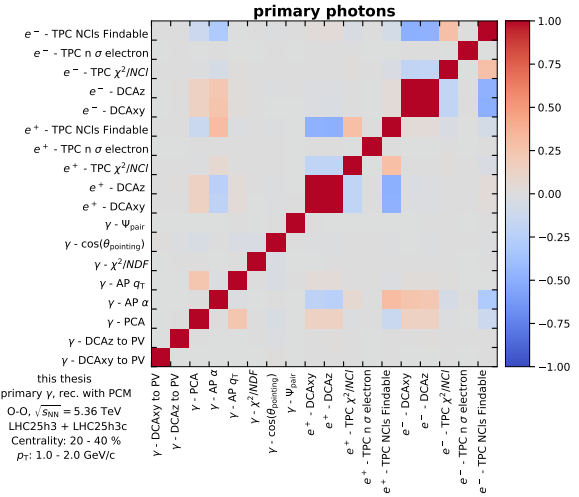
(b) 0.3-0.5 GeV/c



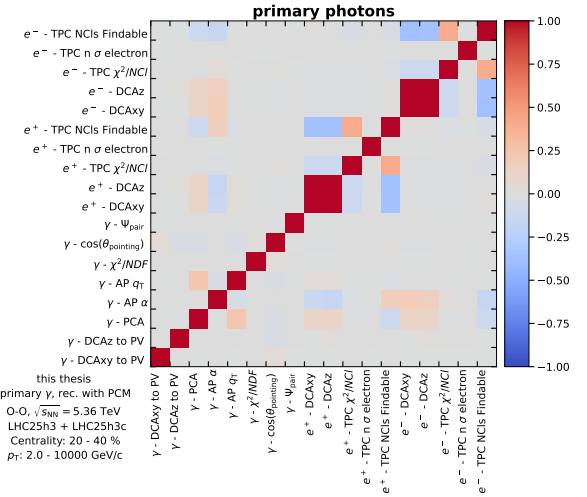
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

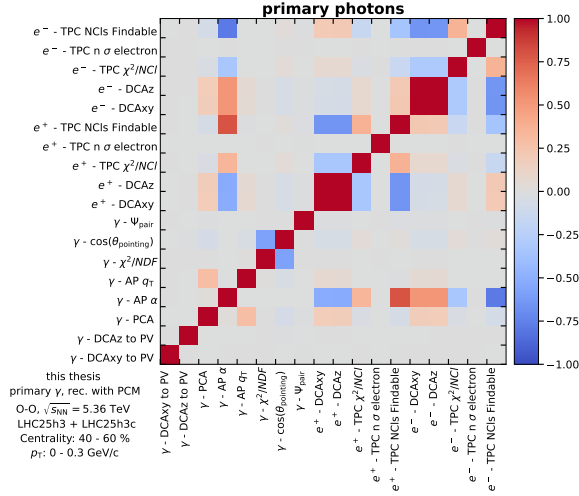


(e) 1.0-2.0 GeV/c

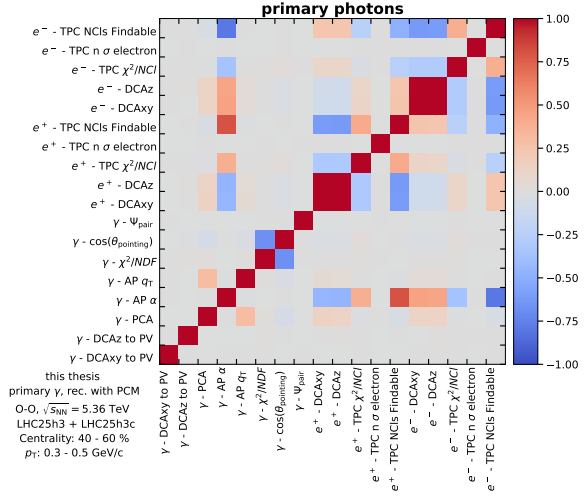


(f) 2.0-10000 GeV/c

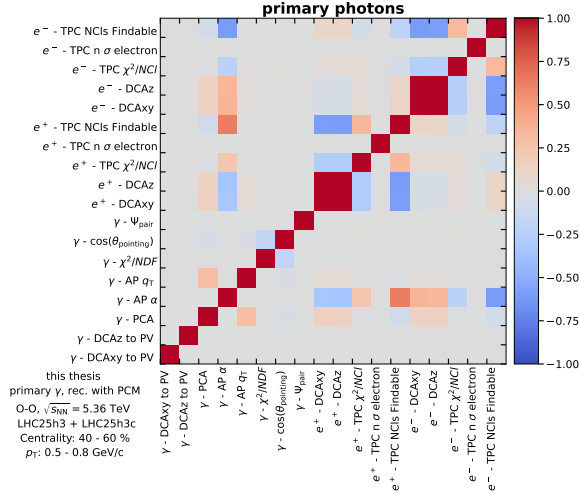
Figure 15: Correlation map of primary photons for all p_T bins in the centrality range 20-40 %



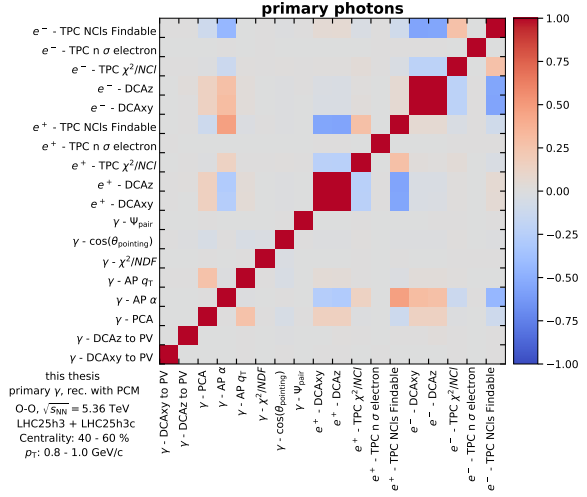
(a) 0-0.3 GeV/c



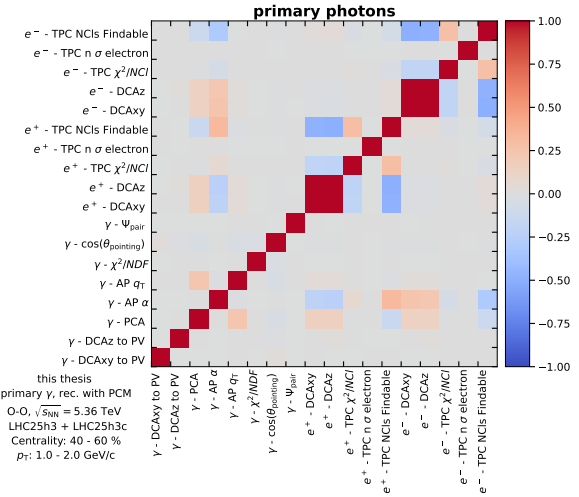
(b) 0.3-0.5 GeV/c



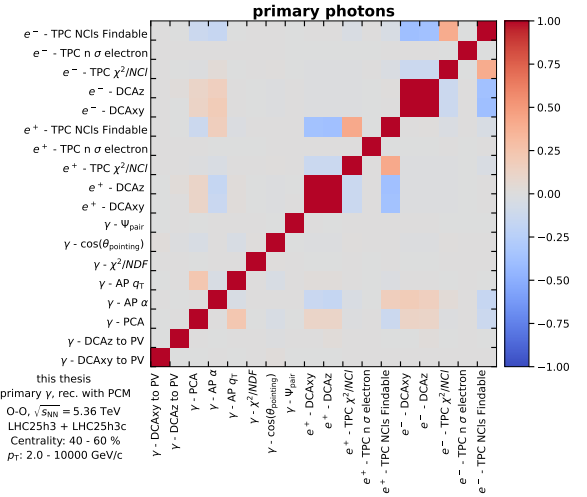
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

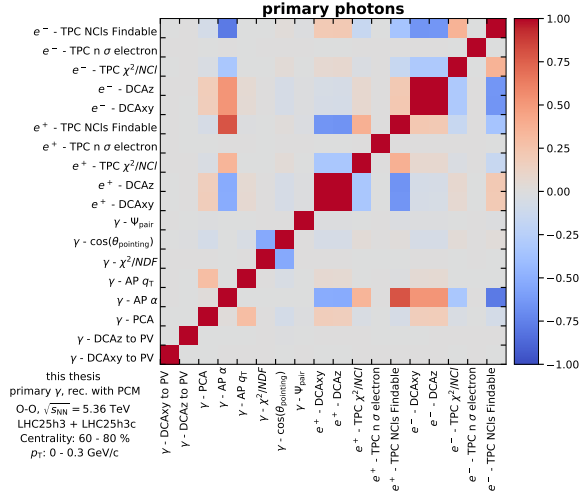


(e) 1.0-2.0 GeV/c

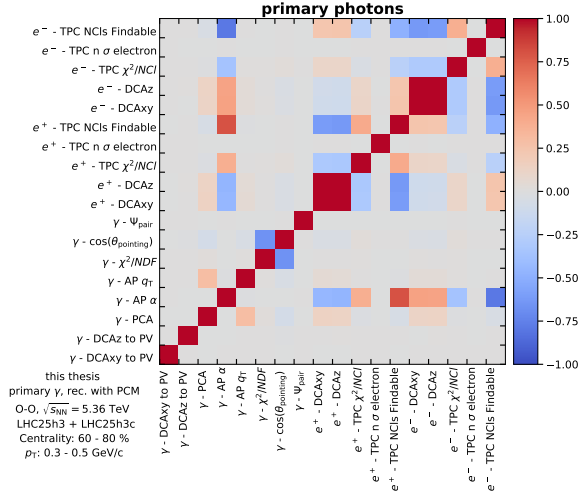


(f) 2.0-10000 GeV/c

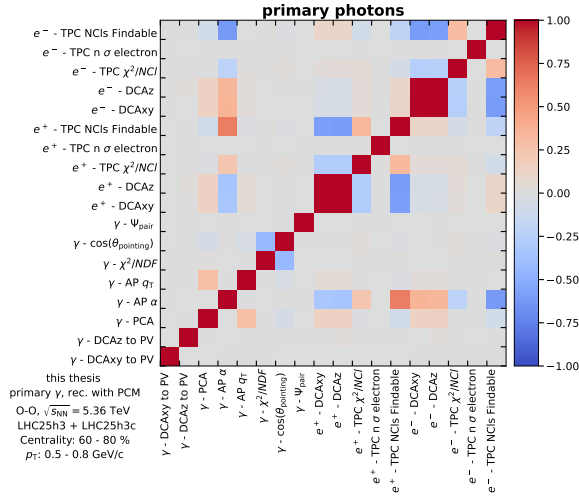
Figure 16: Correlation map of primary photons for all p_T bins in the centrality range 40-60 %



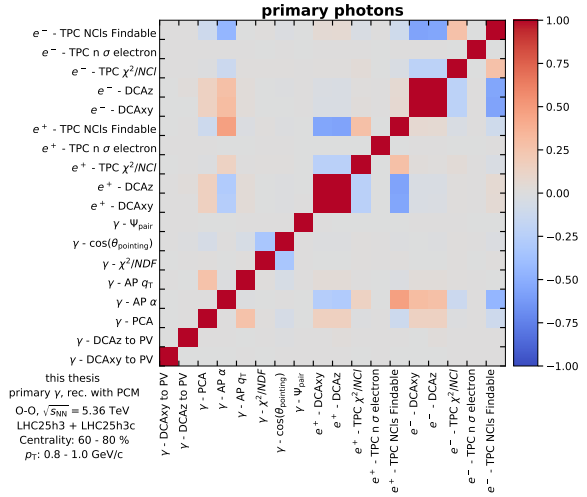
(a) 0-0.3 GeV/c



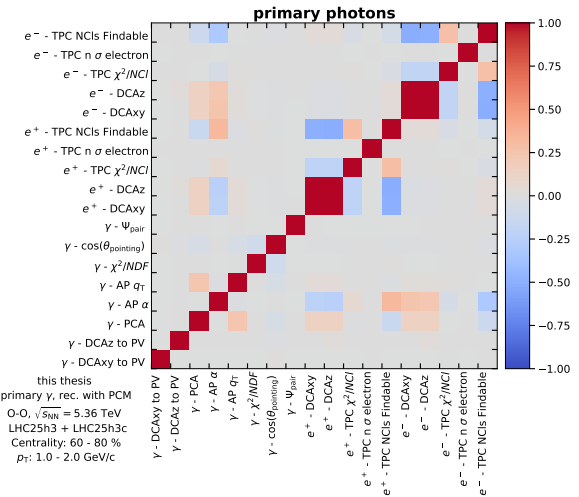
(b) 0.3-0.5 GeV/c



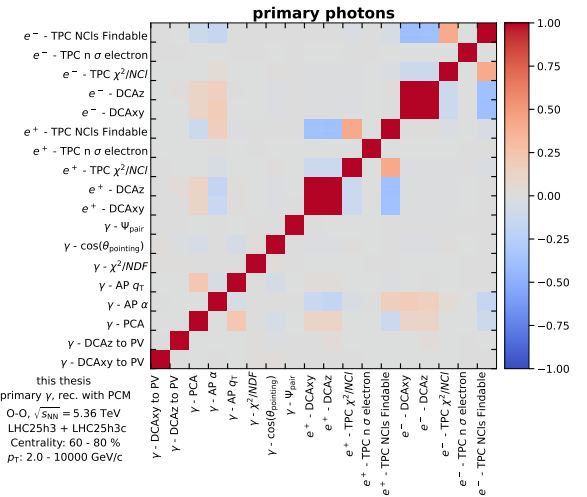
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

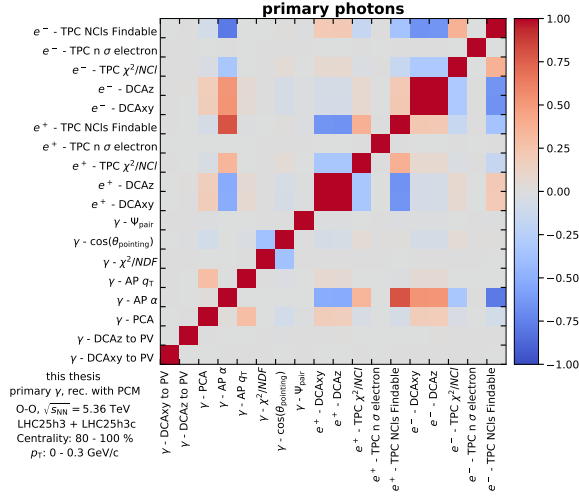


(e) 1.0-2.0 GeV/c

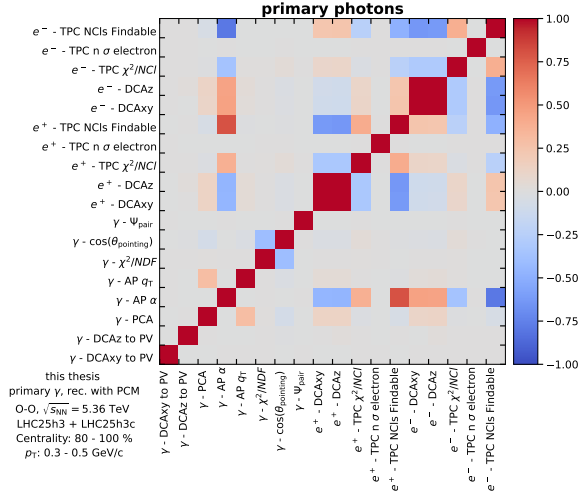


(f) 2.0-10000 GeV/c

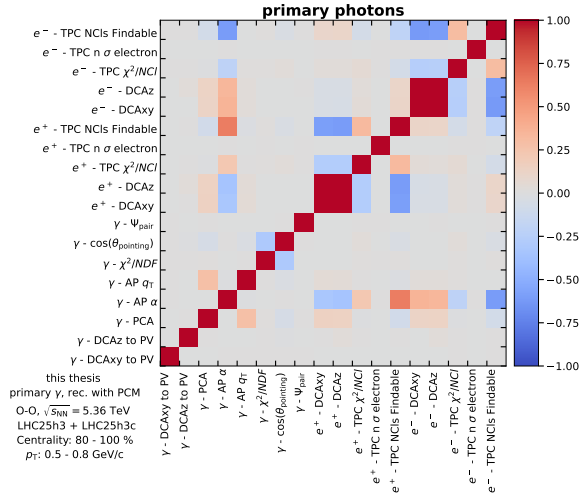
Figure 17: Correlation map of primary photons for all p_T bins in the centrality range 60-80 %



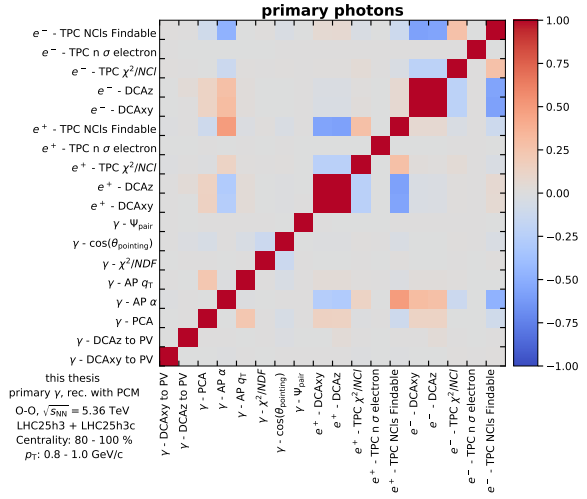
(a) 0-0.3 GeV/c



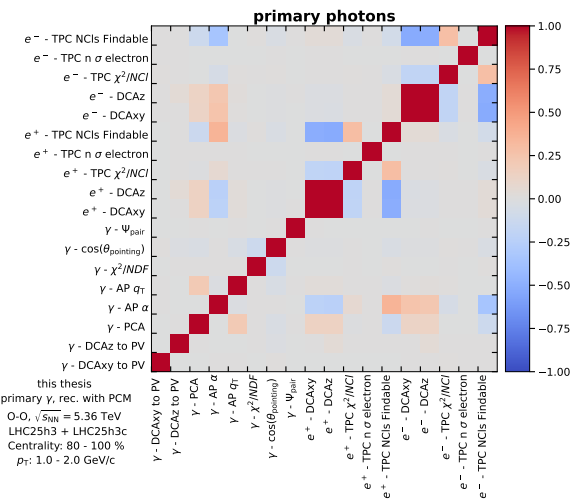
(b) 0.3-0.5 GeV/c



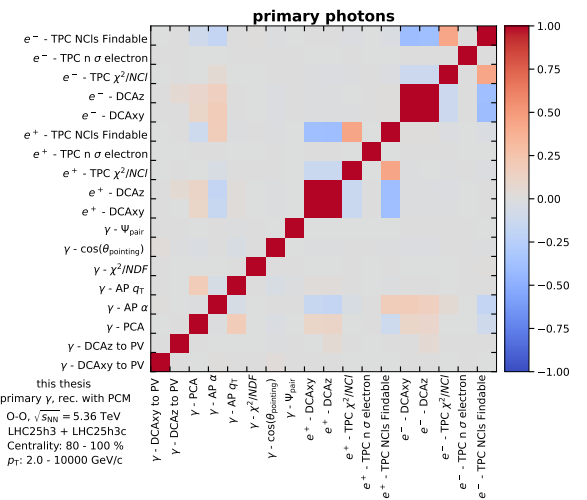
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c



(e) 1.0-2.0 GeV/c



(f) 2.0-10000 GeV/c

Figure 18: Correlation map of primary photons for all p_T bins in the centrality range 80-100 %

3.2 Correlations of background candidates

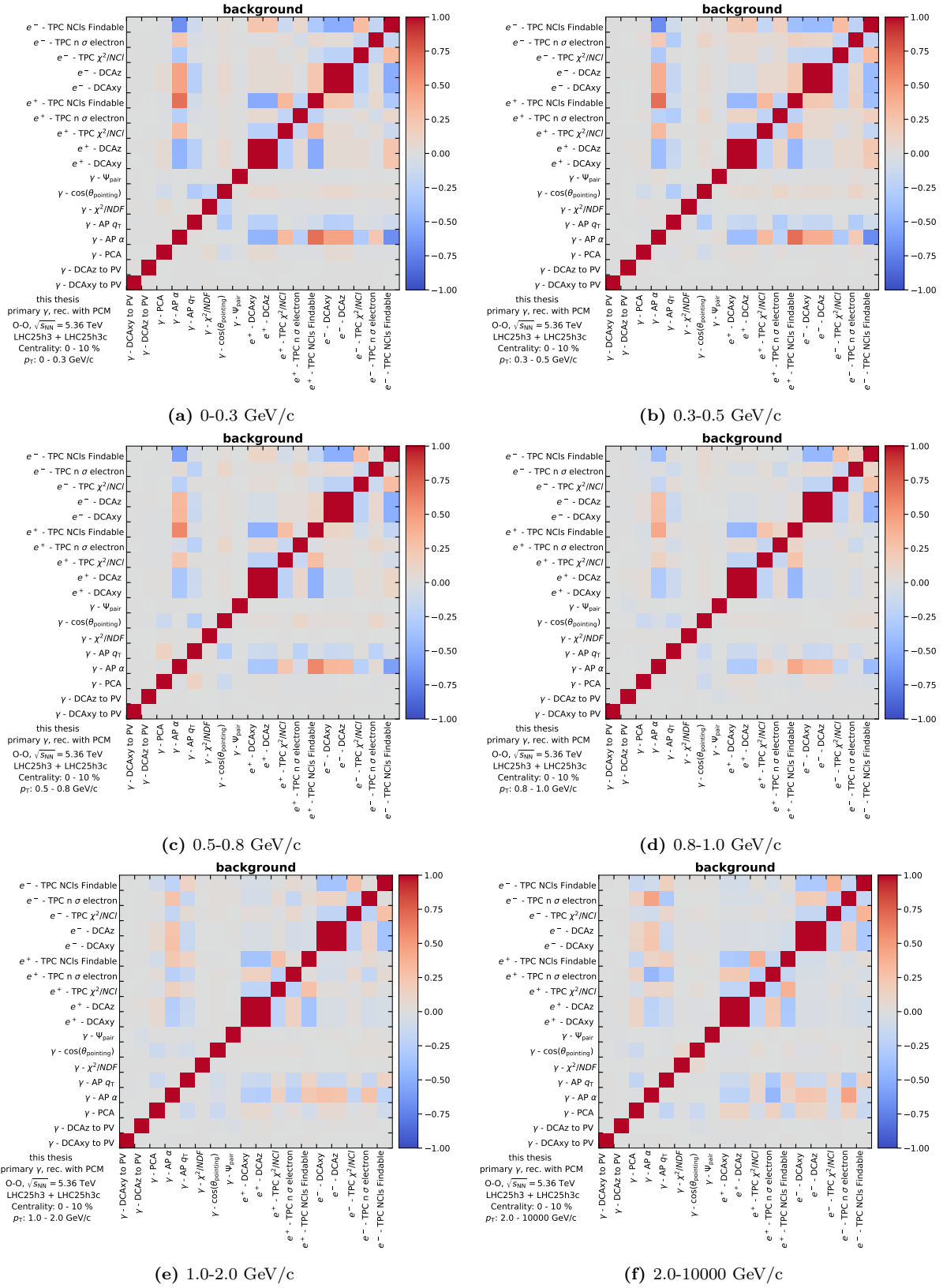
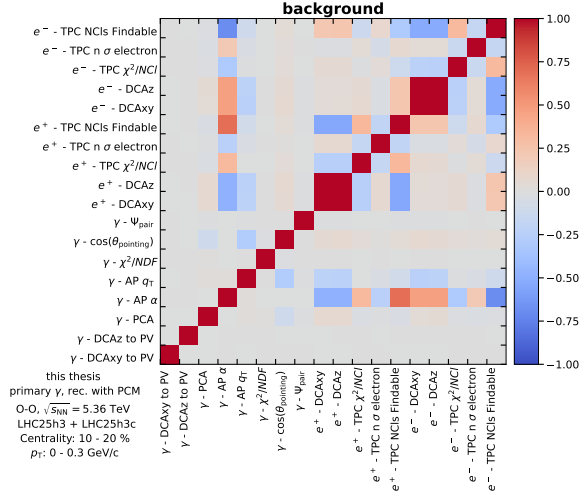
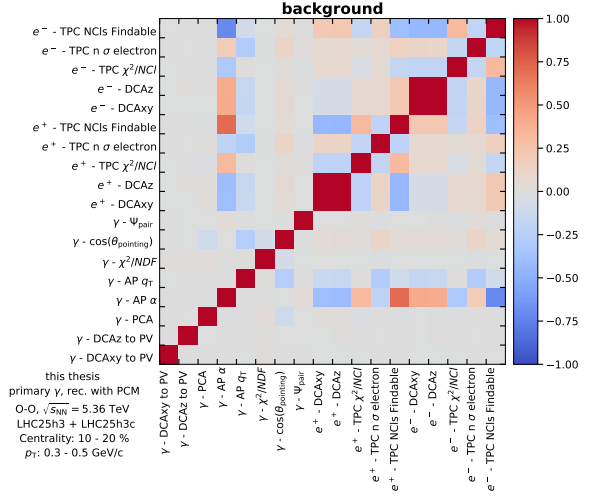


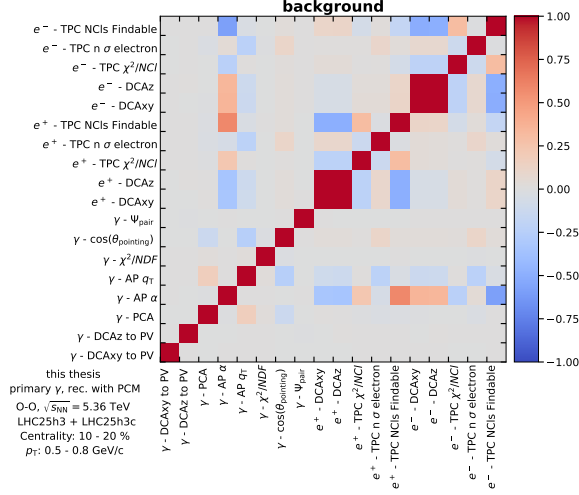
Figure 19: Correlation map of background candidates for all p_T bins in the centrality range 0-10 %



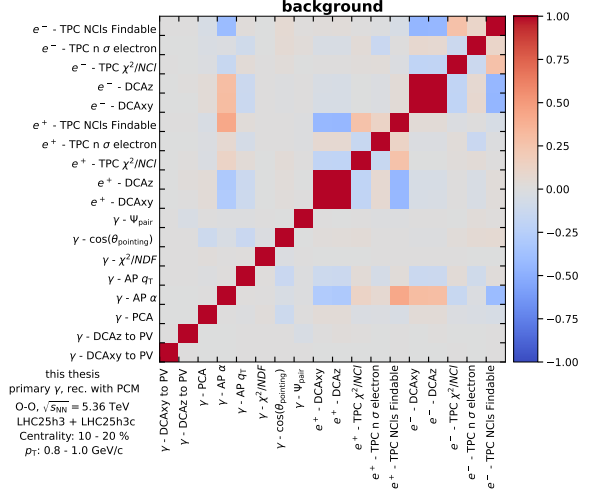
(a) 0-0.3 GeV/c



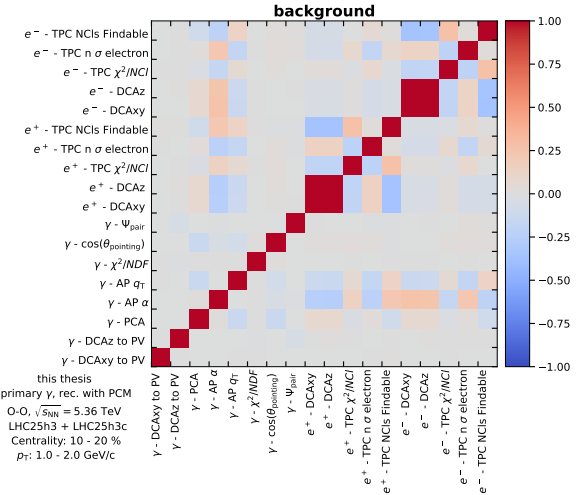
(b) 0.3-0.5 GeV/c



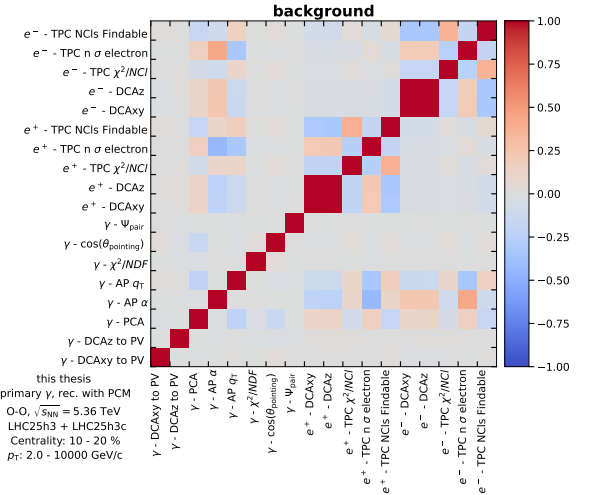
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

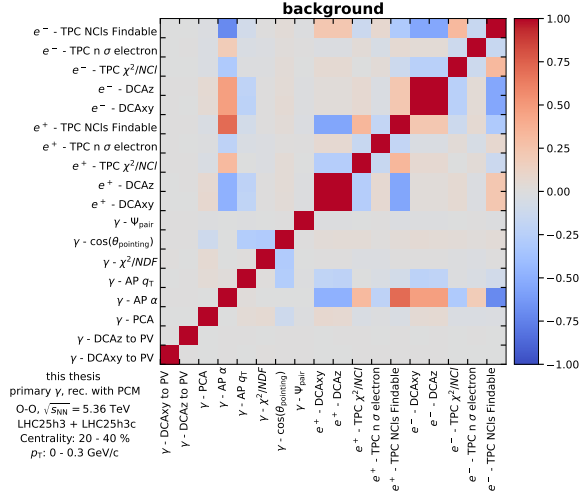


(e) 1.0-2.0 GeV/c

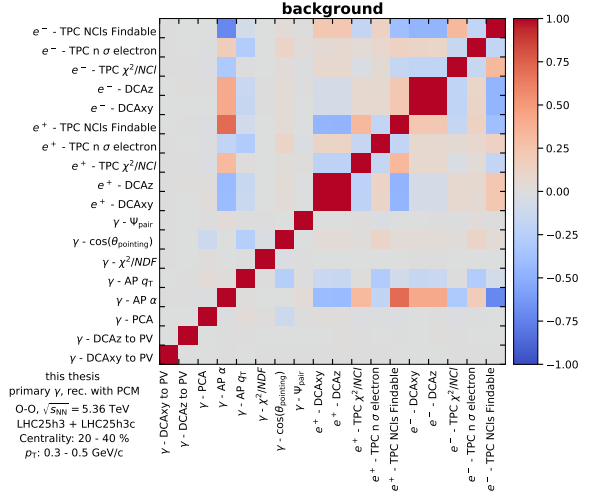


(f) 2.0-10000 GeV/c

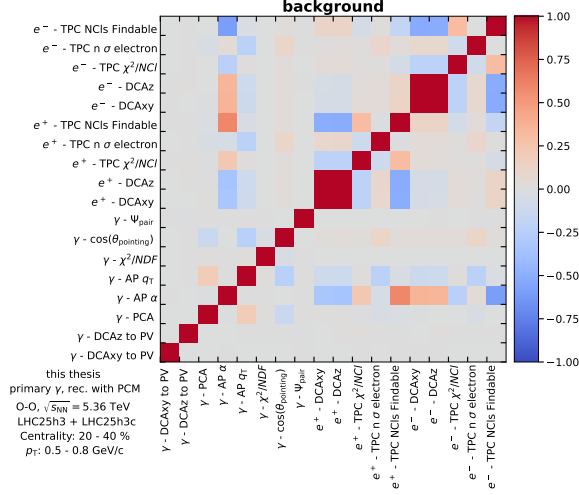
Figure 20: Correlation map of background candidates for all p_T bins in the centrality range 10-20 %



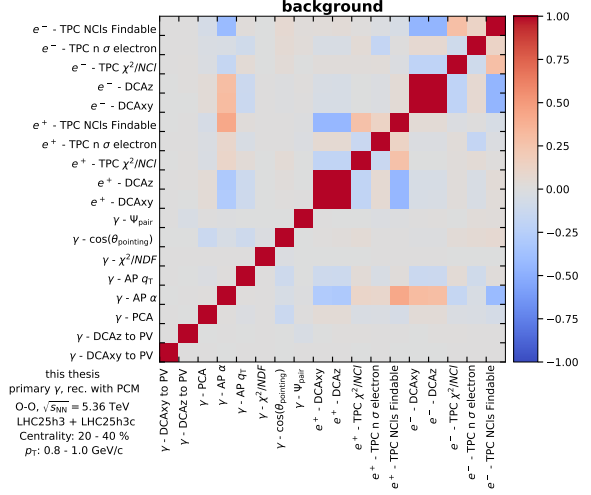
(a) 0-0.3 GeV/c



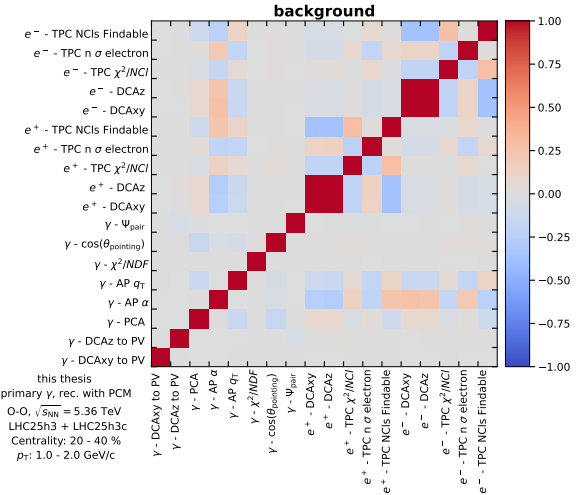
(b) 0.3-0.5 GeV/c



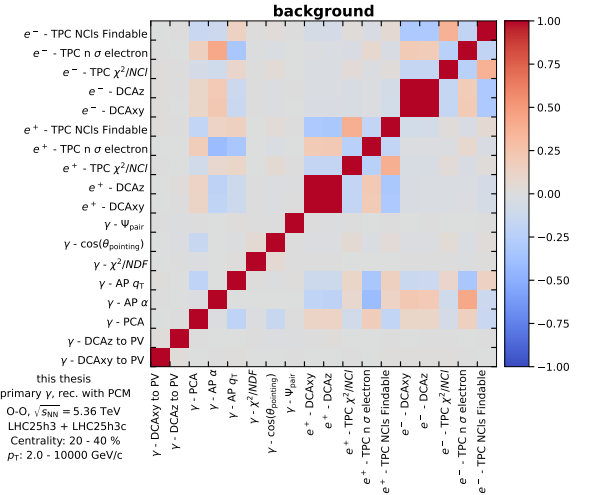
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

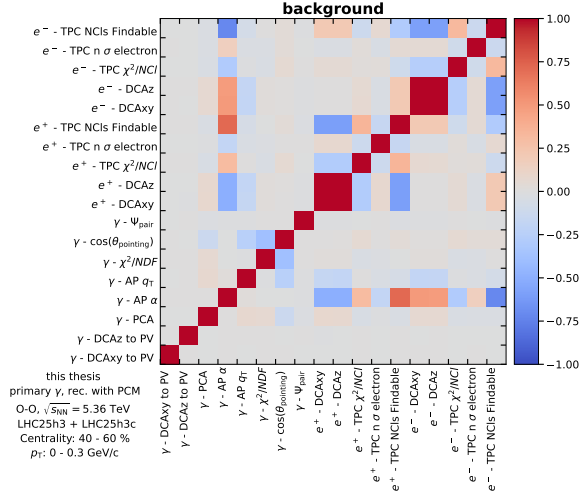


(e) 1.0-2.0 GeV/c

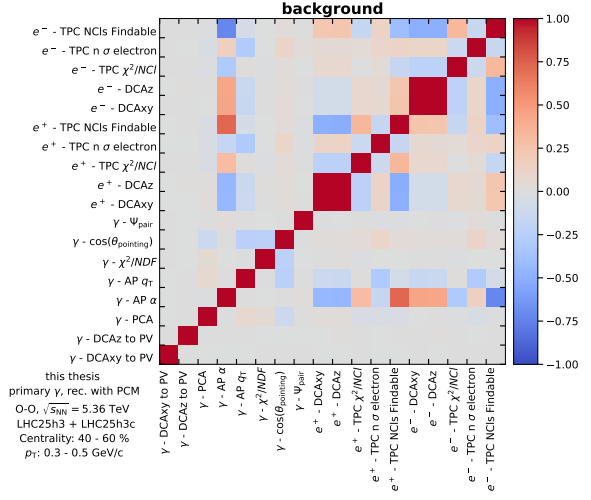


(f) 2.0-10000 GeV/c

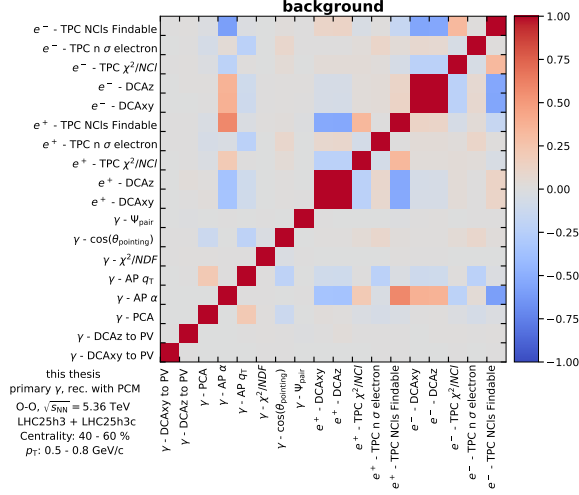
Figure 21: Correlation map of background candidates for all p_T bins in the centrality range 20-40 %



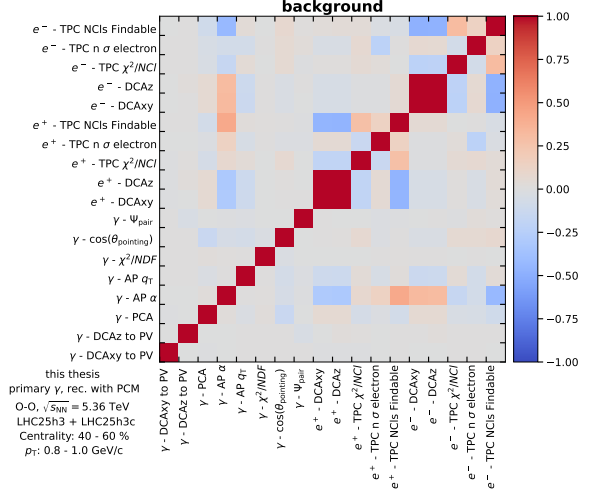
(a) 0-0.3 GeV/c



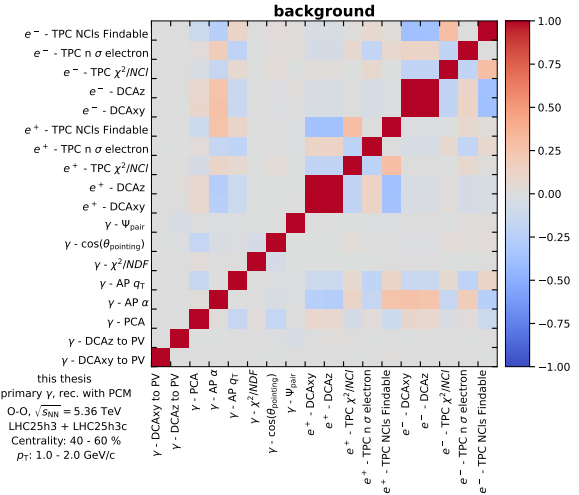
(b) 0.3-0.5 GeV/c



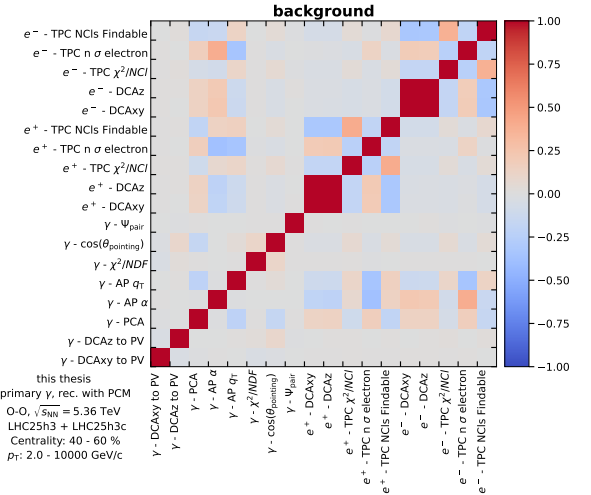
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

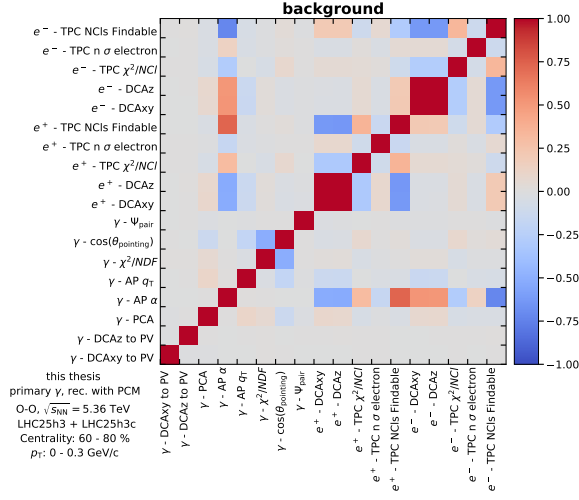


(e) 1.0-2.0 GeV/c

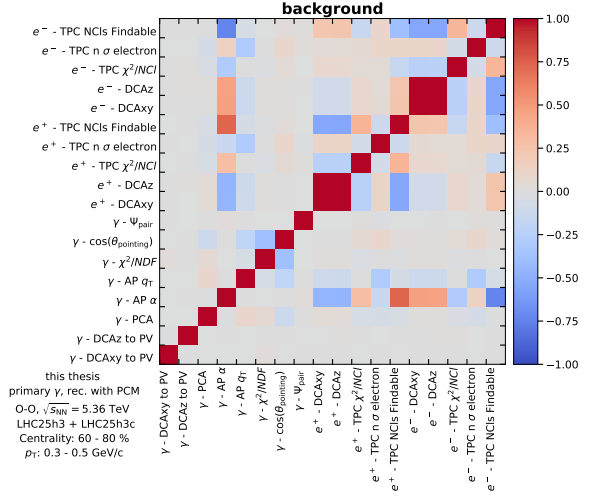


(f) 2.0-10000 GeV/c

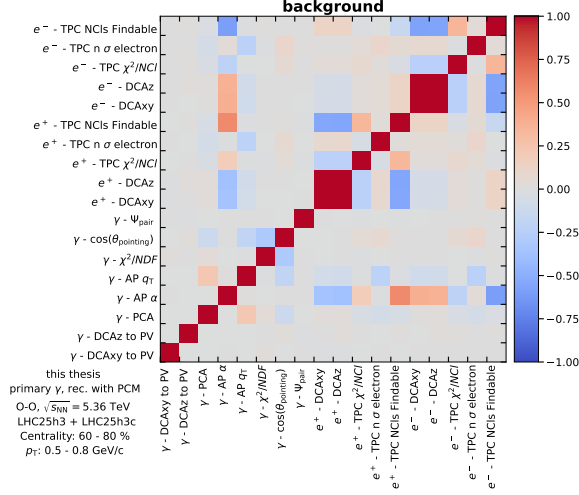
Figure 22: Correlation map of background candidates for all p_T bins in the centrality range 40-60 %



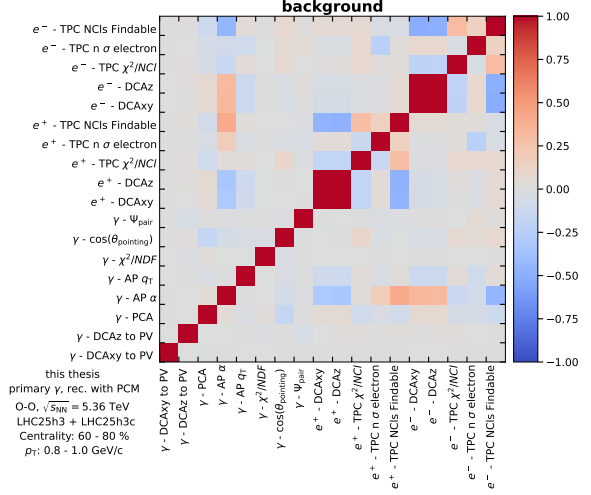
(a) 0-0.3 GeV/c



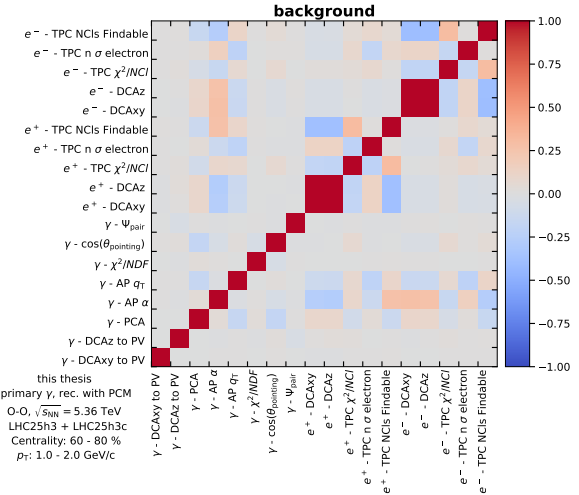
(b) 0.3-0.5 GeV/c



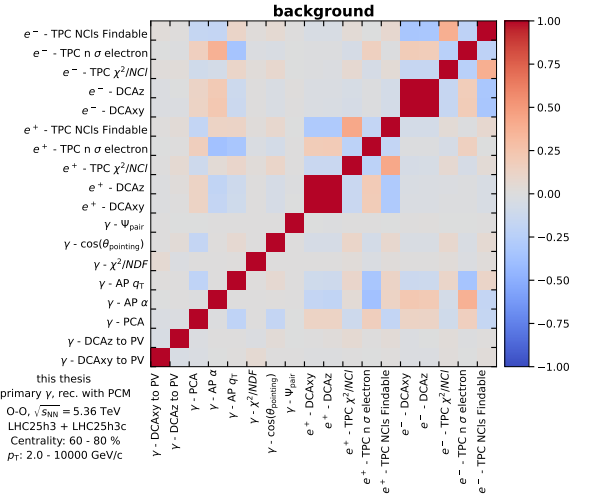
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

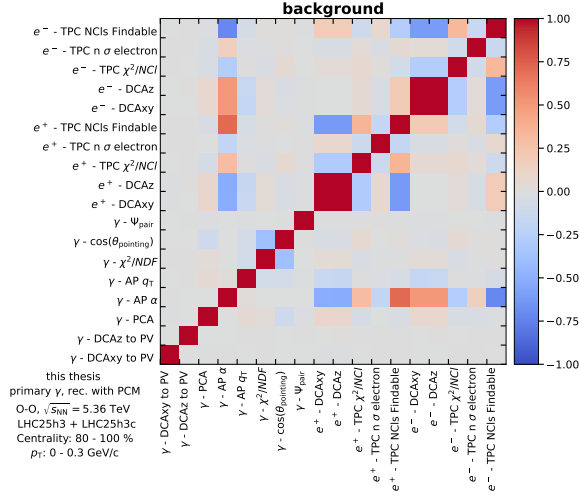


(e) 1.0-2.0 GeV/c

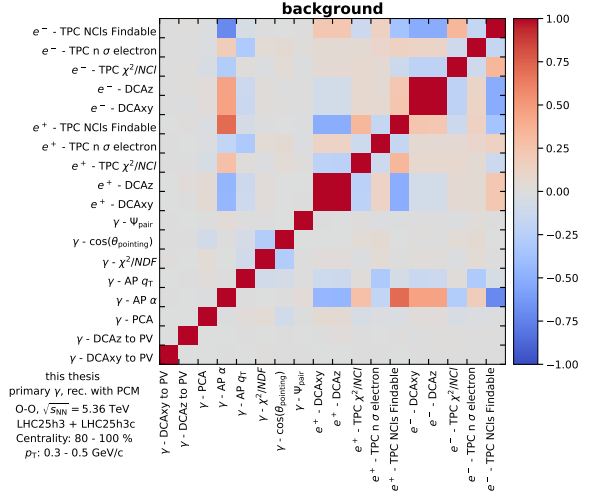


(f) 2.0-10000 GeV/c

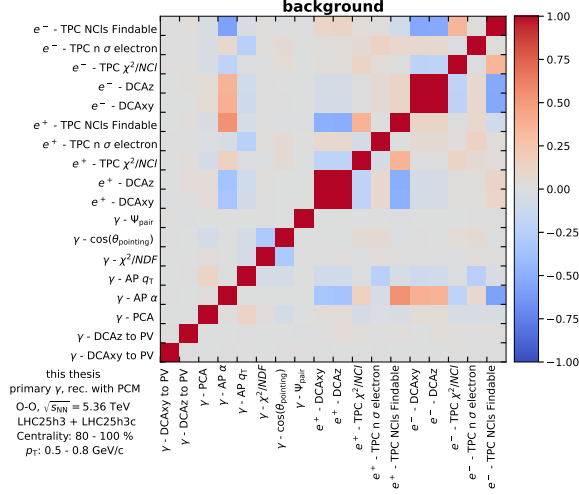
Figure 23: Correlation map of background candidates for all p_T bins in the centrality range 60-80 %



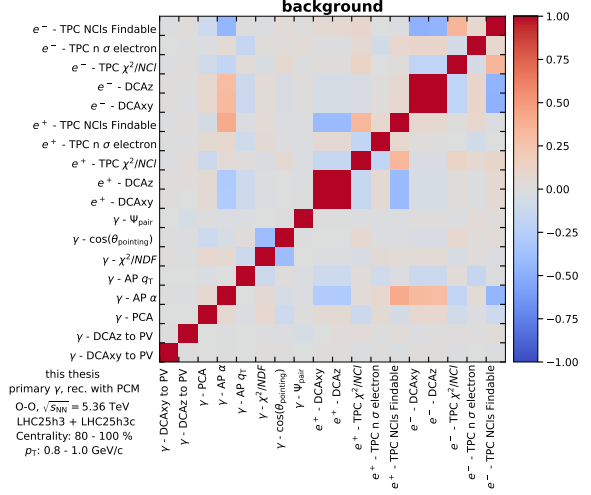
(a) 0-0.3 GeV/c



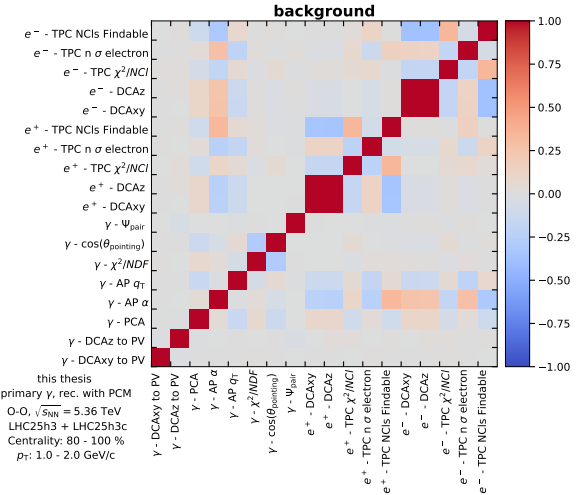
(b) 0.3-0.5 GeV/c



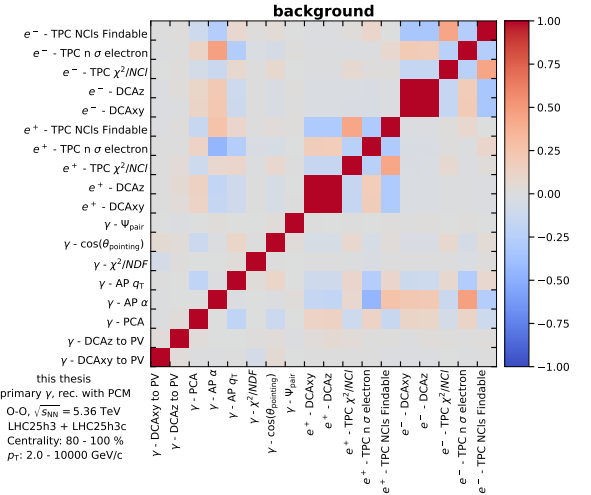
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c



(e) 1.0-2.0 GeV/c



(f) 2.0-10000 GeV/c

Figure 24: Correlation map of background candidates for all p_T bins in the centrality range 80-100 %